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A Meta-Analysis of the Twin Transition: The Impact of Digitalization on Firms' Eco-Innovation

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ABSTRACT

In the twin transition, firms pursue digitalization and environmental sustainability simultaneously, an effort that draws growing academic interest. However, empirical insights into the relationship between digital technologies and eco-innovation remain lacking. With a quantitative synthesis and meta-regression of 38 empirical studies published in high-impact journals, encompassing 587 effect sizes, the current research clarifies the heterogeneous nature of this relationship across technology types and environmental outcomes. Digitalization is positively and statistically significantly associated with eco-innovation. Across technology types, artificial intelligence, digital innovation, and smart product platforms show the strongest positive associations, while robotics and Industry 4.0 technologies yield weaker evidence. Across environmental outcomes, the results indicate robust positive associations with green innovation but negative or null associations on average for decarbonization. In revealing that digitalization alone might not be sufficient to achieve emissions reductions without complementary environmental strategies, this study contributes to twin transition literature and clarifies the heterogeneous nature of the digitalization-eco-innovation relationships. For managers seeking to balance digital investments with environmental objectives, it also offers practical recommendations.

JEL Classification: O31, O32, O33

1 | Introduction

According to the concept of the twin transition, digital and ecological (or green) transformations are not concurrent strategies, but rather function as synergistic forces, capable of reshaping the modern economy (Demirel et al. 2025). In European policy, the European Green Deal frames the green and digital transitions as complementary dimensions of the transformation toward a more climate-neutral and resource-efficient economy, highlighting the potential role of digital technologies in supporting these objectives (European Commission 2019). Theoretical discussions similarly predict that digitalization can enhance organizational capabilities by reducing information asymmetries, optimizing resource flows, and enabling intelligent monitoring (Horbach 2024; Y. Liu et al. 2024). The empirical reality is considerably more complex. A substantial body of research identifies positive synergies between digital transformation and green innovation (Škare et al. 2024; G. Yang et al. 2023), but a prominent counter narrative also indicates negligible or adverse associations, particularly in capital intensive sectors such as oil processing or coal refining (L. Wang, Chen, et al. 2024). Is digitalization positively associated with eco-innovation, or do its documented associations actually reflect specific measurement choices, sectoral contexts, or publication biases?

To address this research, we undertake a meta-regression analysis (MRA) of research conducted on the digitalization-eco-innovation relationships. In accordance with MRA protocols (Havránek et al. 2020; Stanley and Doucouliagos 2014), we synthesize 587 effect sizes at the firm level, drawn from 38 primary studies published in high-impact journals. Beyond averaging these prior results, we apply multiple bias-correction techniques

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that can separate the true empirical effect from any publication bias.

After correcting for publication bias, our results reveal that digital technologies are statistically significantly and positively associated with eco-innovation, but the magnitude of this association is modest. We also identify a critical divergence in outcomes indicating that digitalization is robustly associated with green innovation outputs (e.g., patents, new product development), but its associations with actual decarbonization (CO_2 emission reductions) are statistically weak or negative. Significant heterogeneity also arises across technologies, such as artificial intelligence (AI), digital innovation, and smart product platforms emerge as consistently positively associated with eco-innovation, but industrial robotics and Industry 4.0 technologies yield inconclusive results.

The paper is structured as follows: in Section 2, we establish the key concepts for this research, related to digitalization and eco-innovation, and also note relevant definitions provided in the studies included in the meta-analysis. Section 3 details the methodology and econometric procedures; Section 4 contains the results of the meta-analysis and robustness tests. After discussing theoretical and practical implications in Section 5, we conclude with a discussion of some limitations of this study and suggestions for further research.

2 | Literature Review and Theoretical Background

2.1 | Theoretical Framework

To provide a stronger theoretical basis for understanding the heterogeneous evidence across digital technologies, the analysis draws on the resource-based view (RBV) and the dynamic capabilities view (DCV). From an RBV perspective (Barney 1991), digital technologies can be understood as organizational resources whose contribution to eco-innovation depends on how they are combined with other firm-specific resources and capabilities, consistent with evidence that resource bundle attributes shape value creation in digital firms (Giustiziero et al. 2023). This perspective is particularly relevant because different digital technologies may provide different types of resources and therefore may be associated with eco-innovation through different organizational pathways. The DCV (Teece et al. 1997) complements this perspective by emphasizing firms' ability to develop, integrate, and reconfigure resources and capabilities in response to technological and environmental changes, as evidenced by the positive influence of AI-enabled dynamic capabilities on environmental performance (Fosso Wamba, Queiroz, and Trinchera 2024). Recent firm-level evidence also links corporate digital transformation to eco-innovation, while documenting differences in the associations across digital technologies (Ionaşcu et al. 2026).

The existing literature explicitly connects digital transformation with dynamic capabilities and innovation, including evidence that dynamic capabilities mediate the relationship between digital transformation and low-carbon technology innovation (G. Yang et al. 2023), while other work highlights absorptive capacity and transformative capacity as organizational capabilities involved in the relationship between digital servitization and green innovation (M. J. Lee et al. 2024). Related studies further emphasize the importance of firm-level capabilities and strategic orientation in translating digital resources into innovation outcomes. Digital orientation has been associated with sustainable innovation, particularly in relation to firms' internal and external conditions (Nasiri et al. 2022). More directly, Ranjan (2024) draws on the RBV and its dynamic capability extension to explain how digital orientation translates into innovation performance through complementary digital business capabilities and under different environmental conditions.

This perspective also helps distinguish between the adoption of digital technologies and their association with various forms of eco-innovation. Digital technologies can be linked to changes in products, processes, and organizational practices, yet such changes do not necessarily translate into equivalent improvements in environmental performance or reductions in emissions. From the RBV perspective, this distinction is relevant, as varying environmental outcomes may depend on different combinations of firm-specific and complementary resources (Giustiziero et al. 2023). From the DCV perspective, firms may also differ in their ability to integrate digital resources into existing activities and to adapt or reconfigure those activities in light of environmental objectives. Thus, the same type of digital resource can be associated with different eco-innovation outcomes depending on the technological and organizational context of its implementation. This reasoning is relevant to the distinction between eco-innovation and decarbonization, given that digitalization can be linked to innovation-oriented environmental activities without necessarily entailing equivalent emission reductions.

Taken together, the RBV and DCV provide a conceptual basis for interpreting why the association between digitalization and eco-innovation may vary across technologies and types of eco-innovation, as digital technologies do not constitute homogeneous resources, and firms may differ in their ability to mobilize and reconfigure them for environmental innovation. Accordingly, the RBV and DCV are used in this study not to impose a deterministic causal explanation on the meta-analytic evidence, but to provide a theoretical framework for interpreting the heterogeneous associations identified across the empirical literature.

2.2 | Digitalization

Digitalization can be understood as the incorporation of digital technologies into business processes to improve operational efficiency and support green innovation (D. Zheng et al. 2024), including the adoption of data analytics, cloud computing, mobile technologies, and big data analytics (Hagan et al. 2024). Its potential contribution to corporate green innovation has received growing attention (Tao et al. 2025). As firms attain greater digital maturity, digitalization may evolve into broader digital transformation through the reconfiguration of production processes, business models, control systems, organizational culture, and communication technologies that improve information efficiency (Chen 2022; Verhoef et al. 2021; L. Wang, Chen, et al. 2024; Zhong and Ren 2025). Thus, digitalization extends beyond the adoption of individual technologies to their

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development and deployment within broader organizational structures and processes.

Prior literature consequently conceptualizes digitalization as a multidimensional rather than homogeneous phenomenon. Montresor and Vezzani (2023), for example, distinguish Internet-based technologies, AI applications, and other advanced industrial technologies. Building on these distinctions and the operationalizations identified in the empirical literature, the following six categories are identified: AI, smart product platforms, digital innovation, industrial robotics, Industry 4.0 technologies, and other digital orientations. Table A1 presents the characteristics, definitions, and operationalizations of the variables included in the meta-regression analysis. In the following paragraphs, we address the dimensions of digital technologies.

AI occupies an important position in contemporary technological and industrial transformations because of its capacity to support decision-making and perform tasks traditionally requiring human intelligence (J. Wang et al. 2025). Its organizational relevance has been examined in relation to intelligent behavior, decision-making, and other organizational activities (Di Vaio et al. 2020; J. Liu et al. 2020; Mikalef and Gupta 2021; Tian et al. 2023).

Smart product platforms represent the evolution of smart product service ecosystems from earlier smart product service systems (M. Zheng et al. 2017), with empirical evidence linking such platforms to green technology innovation trajectories (W. Zhang et al. 2024). They reflect business models based on smart technologies and customer interactions, linking digitalization with organizational and market-oriented processes relevant to eco-innovation (M. Zheng et al. 2017).

Digital innovation refers to integrating digital technologies into the discovery and implementation of new digital products and services, improvements in production processes, and the creation or modification of digital business models (Fichman et al. 2014; Hund et al. 2021). It may rely on digital platforms such as e-commerce and online platforms (Dong et al. 2024), with manifestations including e-commerce, intelligent customer service, and the industrial Internet (G. Yang et al. 2023). Thus, this dimension captures both technological adoption and associated organizational and business-model changes.

Industrial robotics represents a digitalization dimension associated with advanced manufacturing (Huang et al. 2026) and intelligence-based industrial production (E. Z. Wang et al. 2022). Industrial robots are multipurpose, reprogrammable, automatically controlled machines whose use can be associated with cleaner production and more eco-efficient resource and material use (L. Wang et al. 2023), thereby reflecting the integration of digital technologies into physical production.

Industry 4.0 technologies encompass technologies associated with the digital transformation of industrial production. Blockchain, for example, has been examined in relation to the quality of innovation in manufacturing firms (Chin et al. 2021). As a distributed system for cryptographically recording and storing consistent, immutable, and linear records of events and transactions, blockchain illustrates how digital technologies can alter information flows and interactions within production and innovation systems (Fosso Wamba et al. 2020).

Finally, other digital orientations capture digitalization from a broader strategic and organizational perspective. Digital strategic orientations emphasize that technological adoption may be accompanied by strategic intent and organizational capabilities (Nasiri et al. 2022). Digital resources embedded in internal structures, including digital governance, may therefore reflect a broader orientation relevant to environmental innovation performance (Bendig et al. 2023; Hao et al. 2024; Ranjan 2024; H. Zhang et al. 2026). Investments in technology similarly reflect organizational commitment to acquiring digital technologies and devices relevant to eco-innovation (Appio et al. 2021; Ye et al. 2023), while technological innovation efficiency captures the ability to transform investments in R&D and human talent into tangible results (Z. Liu et al. 2025).

Overall, these dimensions show that digitalization encompasses heterogeneous technological resources, applications, and organizational orientations rather than a single homogeneous construct. This distinction is conceptually important for the digitalization-eco-innovation relationships because different technologies may provide different informational, operational, organizational, and innovation-related resources, providing a basis for examining whether their association with eco-innovation varies across technology types.

2.3 | Eco-Innovation

According to Kemp and Pearson (2007), eco-innovation involves innovations that reduce environmental risks and pollution. From this perspective, firms engage in eco-innovation when they introduce new production models and practices that reduce environmental risks, pollution, and other negative environmental impacts (Acebo et al. 2021a; Cattani et al. 2025; Montresor and Vezzani 2023) and contribute to greater sustainability (García-Granero et al. 2020). In the context of the twin transition, eco-innovation represents the environmental dimension of a broader transformation in which green and digital technological change interact, potentially contributing to the development of a greener and more digital economy (Demirel et al. 2025). The empirical literature reviewed in this study primarily distinguishes between two dimensions of eco-innovation: green innovation and decarbonization.

Green innovation encompasses environmentally sustainable products, processes, and technologies. Its multidimensional nature is reflected in studies examining digitalization in relation to green products and processes (Ardito et al. 2021; Jibril et al. 2024; Lin et al. 2024) and green technologies (Shao et al. 2024; Xu et al. 2023; W. Zhang et al. 2024). In general, green innovation seeks to reduce energy consumption and pollution, with its environmental contribution depending on the quantity and quality of innovation achieved. This may involve ecological raw materials, reduced material use, lower emissions, and more efficient energy consumption (Lin et al. 2024).

The literature distinguishes several types of green innovation activities, including actions to reduce energy consumption

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and waste emissions, decrease the environmental impact of products or services, and implement voluntary environmental practices (M. J. Lee et al. 2024). Other studies distinguish green product and process innovation (Shahzad et al. 2026; Yuan and Cao 2022). Green process innovation reflects firms' ability to develop technologies that improve operational processes (Ranjan 2024), whereas green product innovation may involve nontoxic compounds, biodegradable materials, and product-design changes that reduce environmental impacts and improve energy efficiency (Yuan and Cao 2022). Ecological raw materials also represent an important innovation area, as new materials may coexist with existing ones, such as petroleum-derived and biodegradable plastics (D'Agostino and Moreno 2019), while potentially avoiding additional resource consumption (Ye et al. 2023).

In manufacturing, green innovation can contribute to sustainable economic development (C. C. Lee et al. 2022), while firms may adopt digitalization through green technologies while pursuing economic benefits (S. Wang et al. 2018). This transition also involves organizational and human dimensions, and smart manufacturing transformation has been identified as an important strategic prerequisite for green innovation (H. Yang et al. 2022). Green innovation therefore encompasses not only technological outputs but also changes in organizational processes, managerial orientation, and production practices.

The second dimension of eco-innovation is decarbonization, and it refers to the reduction of environmental emissions. Consistent with Horbach's (2024) definition of eco-innovation as innovation associated with reduced emissions or more efficient resource use, studies operationalize eco-innovation through emissions and environmental-performance indicators (Dong et al. 2024; Huang et al. 2026; D. Zheng et al. 2024; Q. Zhu et al. 2023). Decarbonization is particularly relevant to the twin transition because greenhouse gases can remain in the atmosphere for centuries and contribute to heat accumulation (Filonchyk et al. 2024). Digital technologies associated with reductions in greenhouse gas emissions may therefore contribute to addressing climate change (Y. Liu et al. 2024).

Carbon dioxide accounts for approximately 76% of total greenhouse gas emissions, making decarbonization a primary environmental objective and highlighting the importance of measuring firms' actual carbon emissions (Ding et al. 2024). Traditional measures include carbon emissions per unit of gross domestic product (L. Wang and Ramsey 2024), reflecting the resource waste and emissions associated with industrial production processes. More recent approaches emphasize information and monitoring: carbon sustainability reports can support environmental monitoring and performance assessment (Sharaf-Addin 2024), particularly when supported by digitalization (M. Zhu et al. 2026).

Green innovation encompasses environmentally oriented products, processes, technologies, and organizational changes, whereas decarbonization focuses more directly on emissions and carbon-related environmental performance. This distinction is relevant because digital technologies may be associated with innovation activities without necessarily generating equivalent environmental outcomes, providing a basis for examining the heterogeneous evidence in the literature.

2.4 | Linking Digitalization and Eco-Innovation

The relationship between digitalization and eco-innovation has attracted increasing attention, although a conclusive causal relationship has not been established (Cattani et al. 2025). Existing evidence indicates a positive association between the two constructs in several empirical settings (Montresor and Vezzani 2023; Tian et al. 2023). This perspective is consistent with the view that digital transformation involves broader reconfigurations of production processes, business models, control systems, and organizational culture rather than the isolated adoption of individual technologies (B. Wang, Gong, and Yang 2024; Zhong and Ren 2025).

Digitalization has been associated with improvements in green innovation, resource and energy efficiency, and recycling, although these associations vary across geographical and sectoral contexts (Siedschlag et al. 2024; K. Zhang and Deborah 2025). Digital transformation has also been associated with green technologies, internal management control, and corporate environmental performance, with environmental awareness among senior management potentially reinforcing these associations (Martínez Falcó et al. 2025; Song et al. 2024). Similarly, digital transformation has been associated with production and business efficiency, carbon emissions, corporate pollution, and carbon-emissions performance (Kong and Feng 2025; D. Zheng et al. 2024; Zhou and Liu 2024; Q. Zhu et al. 2023). Other studies report positive associations with low-carbon innovation through technological innovation, environmental problem-solving capabilities, and knowledge capture (Jibril et al. 2024; Shao et al. 2024; B. Wang, Gong, and Yang 2024).

Evidence also links specific technologies with sustainable or green innovation, including robotics (L. Wang et al. 2023), AI (Fosso Wamba, Queiroz, Pappas, et al., 2024), blockchain (Chin et al. 2021), and smart product platforms (W. Zhang et al. 2024). More broadly, digital transformation may be associated with firms' ability to detect market demand and capitalize on opportunities, as well as with improvements in the quality and quantity of green innovation (M. J. Lee et al. 2024; Y. Yang and Luo 2025). In manufacturing, Industry 4.0 provides another pathway associated with sustainable production (Mondal et al. 2024).

These associations, however, are not uniform. Digitalization has been associated with differences in carbon intensity across geographical locations (D. Zheng et al. 2024), while the relationship between industrial robots and emissions varies according to the production process and may be negligible in some settings (Huang et al. 2026). Such evidence suggests that digital technologies alone may not be sufficient to produce uniform eco-innovation outcomes and that organizational context, strategic orientation, technological characteristics, and the way digital resources are deployed may contribute to observed differences.

The distinction between green innovation and decarbonization provides a further basis for this heterogeneity. Digital technologies may be associated with firms' capacity to develop environmentally oriented products, processes, and technologies

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without being associated to the same extent with reductions in actual emissions. The former concerns innovation activities and organizational capabilities, whereas the latter reflects a more direct environmental outcome. Thus, the digitalization-eco-innovation relationship may vary not only across technologies but also across types of eco-innovation, providing a rationale for examining technology- and outcome-specific differences through meta-regression.

3 | Data and Methodology

3.1 | Data Collection

By applying meta-analysis techniques, we identify average effects, as well as publication bias and heterogeneity in the results (Havránek et al. 2020; Irsova et al. 2024; Opatrny et al. 2025; Stanley and Doucouliagos 2014). With a systematic literature review, we identify academic contributions that relate digitalization with eco-innovation in business contexts (Tranfield et al. 2003). In detail, we searched the Web of Science (WoS) database, through the end of December 2025.

Based on the search results, we identified keywords and synonyms from relevant articles (Table A2 in appendix reports the search strings) and applied inclusion and exclusion criteria following the PRISMA approach (Figure 1). We required studies to concern business sectors, be conducted at the firm level, and be published in English in top journals according to the Academic Journal Guide. Table A3 reports the number of articles by journal and country. We excluded qualitative studies, systematic literature reviews, gray literature, and articles unrelated to digitalization or eco-innovation (see Table A4 in appendix for examples).

These criteria reduced the initial keyword-search results to 38 empirical studies reporting 587 effect sizes. Figure 1 details the final sample selection following the PRISMA 2020 format (Page et al. 2021). From these studies, we identified independent variables related to digitalization and dependent variables related to eco-innovation; Table A5 summarizes the resulting data set.

As discussed in the Literature Review and Theoretical Background, digitalization variables are classified according to the conceptual approaches identified in the prior literature.

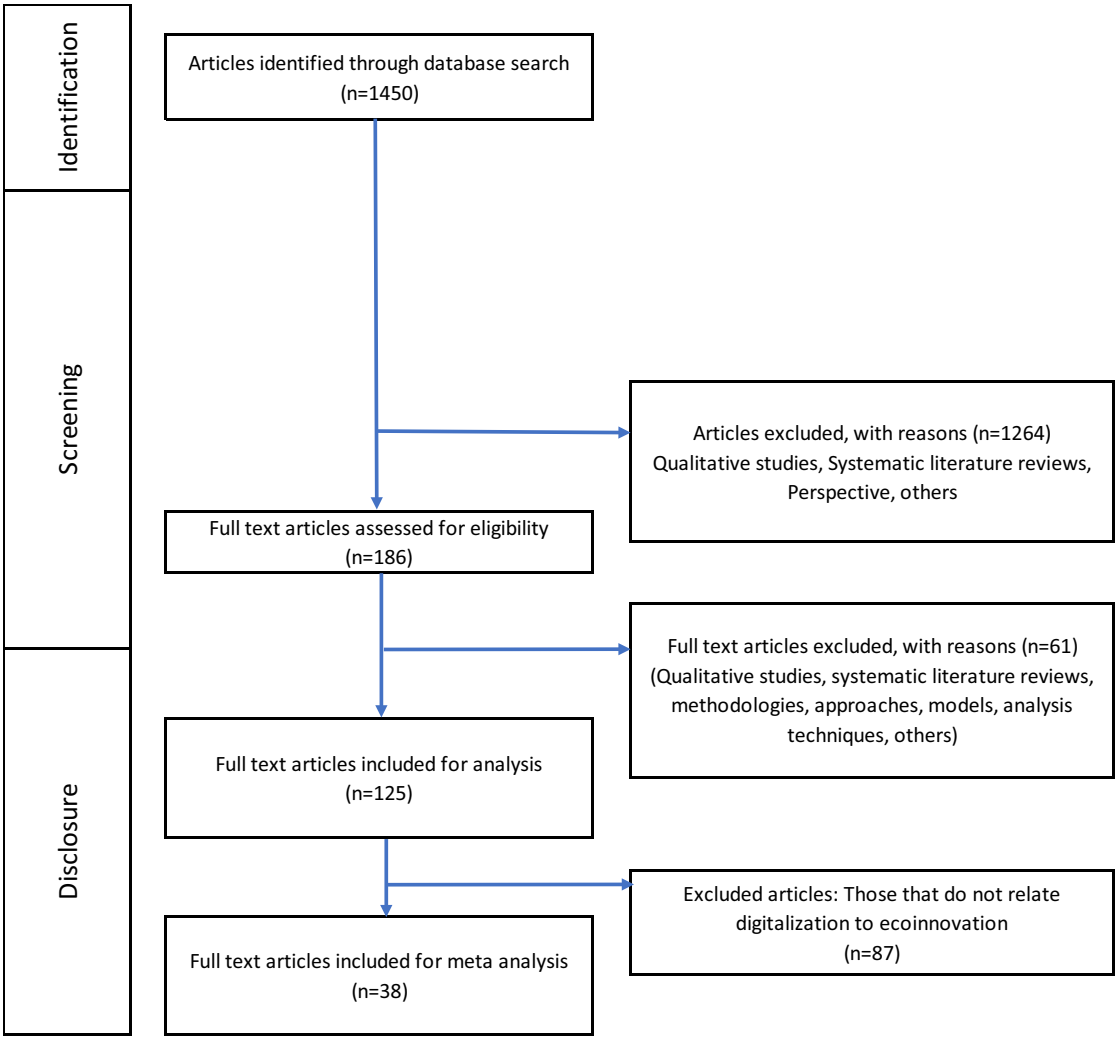


FIGURE 1 | Literature search flow diagram (PRISMA). Flow diagram of the literature search and screening process.

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TABLE 1 | Variables, means, and standard errors.

Variable	Description	Mean	SE
PCC	Partial correlation coefficient	0.046	0.004
Type of digitalization			
Artificial intelligence	=1, if estimate uses data from artificial intelligence, 0 otherwise	0.091	0.015
Digital innovation	=1, if estimate uses data from digital innovation, 0 otherwise	0.021	0.005
Industrial robotics	=1, if estimate uses data from industrial robotics, 0 otherwise	0.002	0.003
Industry 4.0 technologies	=1, if estimate uses data from Industry 4.0 technologies, 0 otherwise	0.059	0.029
Other digital orientations	=1, if estimate uses data from other digital orientations, 0 otherwise	0.095	0.010
Smart product platform	=1, if estimate uses data from smart product platform, 0 otherwise	0.141	0.024
Regional effects			
China	=1, if estimate uses data from China, 0 otherwise	0.037	0.004
Ghana	=1, if estimate uses data from Ghana, 0 otherwise	0.487	0.012
India	=1, if estimate uses data from India, 0 otherwise	0.264	0.008
Italy	=1, if estimate uses data from Italy, 0 otherwise	0.017	0.005
Pakistan	=1, if estimate uses data from Pakistan, 0 otherwise	0.223	0.039
United Kingdom	=1, if estimate uses data from United Kingdom, 0 otherwise	0.022	0.007
USA	=1, if estimate uses data from United States, 0 otherwise	0.152	0.023
Various	=1, if estimate uses data from various nations, 0 otherwise	0.111	0.055
Estimation characteristics			
Control gas emissions	=1, if estimate controls for firm gas emissions, 0 otherwise	0.017	0.005
Control R&D	=1, if estimate controls for firm R&D activities, 0 otherwise	0.055	0.006
Environmental regulation	=1, if estimate controls for firm environmental regulation, 0 otherwise	0.075	0.008
External collaboration	=1, if estimate controls for firm external collaboration, 0 otherwise	0.042	0.012
Control firm exporting	=1, if estimate controls for firm exporting activities, 0 otherwise	0.012	0.003
Control ownership type	=1, if estimate controls for firm ownership type, 0 otherwise	0.046	0.004
Control external financing	=1, if estimate controls for firm external financing activities, 0 otherwise	0.019	0.004
Control firm growth	=1, if estimate controls for firm growth, 0 otherwise	0.025	0.008
Control age	=1, if estimate controls for firm age, 0 otherwise	0.034	0.006
Control size	=1, if estimate controls for firm size, 0 otherwise	0.037	0.005
Model type			
Probit or Logit	=1, if probit or logit method is used for the estimation, 0 otherwise	0.072	0.013
PLS SEM	=1, if structural equation modeling is used for the estimation, 0 otherwise	0.328	0.035
OLS	=1, if ordinary least squares method is used for the estimation, 0 otherwise	0.068	0.009
Instrumental variable method	=1, if instrumental variables method is used for the estimation, 0 otherwise	0.027	0.005
Panel data with fixed effects	=1, if panel data with fixed effects is used for the estimation, 0 otherwise	0.012	0.005
Data characteristics			
Longitudinal data	=1, if longitudinal data is used for the estimation, 0 otherwise	0.031	0.004
Cross sectional data	=1, if study uses cross sectional to estimate, 0 otherwise	0.198	0.025

Note: Mean values correspond to the average value of the reported estimates. All variables were defined and operationalized following previous literature and in accordance with the methodological requirements of this study. Partial correlation coefficient is used as an effect size and results from the transformation of the regression coefficients.

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Table 1 presents this classification and the corresponding descriptive statistics, while Table A6 reports descriptive statistics for the weighted and unweighted variables. Eco-innovation variables were classified into two categories, as noted previously: green innovation and decarbonization.

To begin the analysis, we assigned greater weight to more precise studies and calculated an average effect size across studies. To assess potential publication bias, defined as a tendency toward larger effects in less precise studies, we applied the funnel asymmetry test (FAT, β_1) to detect asymmetry, the precision effect test (PET, β_0) to estimate the true effect, and the precision effect estimate with standard errors (PEESEs). We then accounted for cross-study differences in effect sizes while controlling for publication bias. MRA was selected because it quantifies both the average effect and the influence of moderators and potential effects, including those identified in small studies.

3.2 | Coding Method

We applied existing guidelines for conducting MRA (Irsova et al. 2024; Stanley and Doucouliagos 2014), in terms of extracting and classifying information from the data. A dataset was constructed based on the extraction and compilation of the relevant information and variables reported in the empirical literature reviewed. If studies report only *t*-statistics or *p* values, we calculated the standard errors from these quantities, by dividing the effect size by the *t* values. If they did not report standard errors (SE_{effect}) but indicated the level of statistical significance, we approximated the missing SE_{effect} values by applying the standard relation between test statistics and coefficients, then inferring SEs using the formula $SE_{effect} \approx effect / z$, where z corresponds to the conventional critical values for two-tailed tests. The publication bias tests feature both linear and nonlinear regression approaches, which include the FAT-PET and PEESE estimators, as well as study-level FE and IV methods. As more robust approaches, we also employed the WAAP model and EK regression. For even greater certainty, we confirmed the consistency of the results across the multiple estimators (FAT-PET, PEESE, WAAP, and EK), which helps establish the robustness and transparency of our meta-analysis findings.

3.3 | Conversion to a Common Effect Size

Since regression coefficients reported in primary studies are often not directly comparable quantitatively, effect sizes collected from the literature were transformed into partial correlation coefficients (PCCs). The PCC measures the strength and direction of the association between two variables while holding the influence of other variables constant, thus providing a ceteris paribus measure of association. Its main advantage is that, as a dimensionless measure, it allows for direct comparability of effect sizes across studies included in the meta-analysis. The PCC (r) is calculated using the following mathematical relationship:

$$r = \frac{t}{\sqrt{t^2 + \mathrm{df}}} \tag{1}$$

$$SE\, r = \sqrt{\frac{1 - r^2}{\mathrm{df}}} \tag{2}$$

where t represents the statistical value t on the effect of the relationship, df are the degrees of freedom calculated from the data recorded in the included studies (Stanley and Doucouliagos 2014).

4 | Results

4.1 | Basic MRA

Table 2 presents the weighted and unweighted averages of the PCCs capturing the relationship between digitalization and eco-innovation, based on 587 estimates from 38 studies. The average magnitude is positive but relatively small. Fixed-effects estimates (FEE), which assign greater weight to more precise studies through inverse variance weighting, report a statistically significant average PCC of 0.008. By contrast, unrestricted weighted least squares (WLS) yields a similar but statistically insignificant average effect. The random-effects model (REE), which allows effect sizes to vary across studies due to methodological differences, reports a positive and statistically significant effect of 0.042. This divergence across estimators suggests heterogeneity in the literature and indicates that less precise studies tend to report larger effects. Overall, the relationship is positive but modest and sensitive to the estimation method.

4.2 | Publication Bias

Publication bias occurs because studies with positive and significant effects are more likely to be published. Figure 2 contains a forest plot, illustrating the estimated effects of digitalization on

TABLE 2 | Weighted and unweighted effect size averages.

Partial correlation coefficient	(1) FEE	(2) REE	(3) WLS
Average	0.008***	0.042***	0.008
Standard error	(0.000)	(0.004)	(0.006)
95% confidence interval	[0.008 to 0.008]	[0.034 to 0.049]	[−0.003 to 0.019]

Note: Columns (1–3) report the overall average for PCCs. FEE, REE, and WLS denote fixed effects, random effects, and unrestricted weighted least squares estimates, respectively. Fixed-effects estimates correspond to the inverse-variance weighted mean, with weights equal to $1/SE^2$. The weighted least squares (WLS) estimator is obtained from an intercept-only regression using the specified weights, where the constant term represents the weighted average of the effect sizes. The total number of estimates is 587, obtained from 38 studies.
Statistical significance: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

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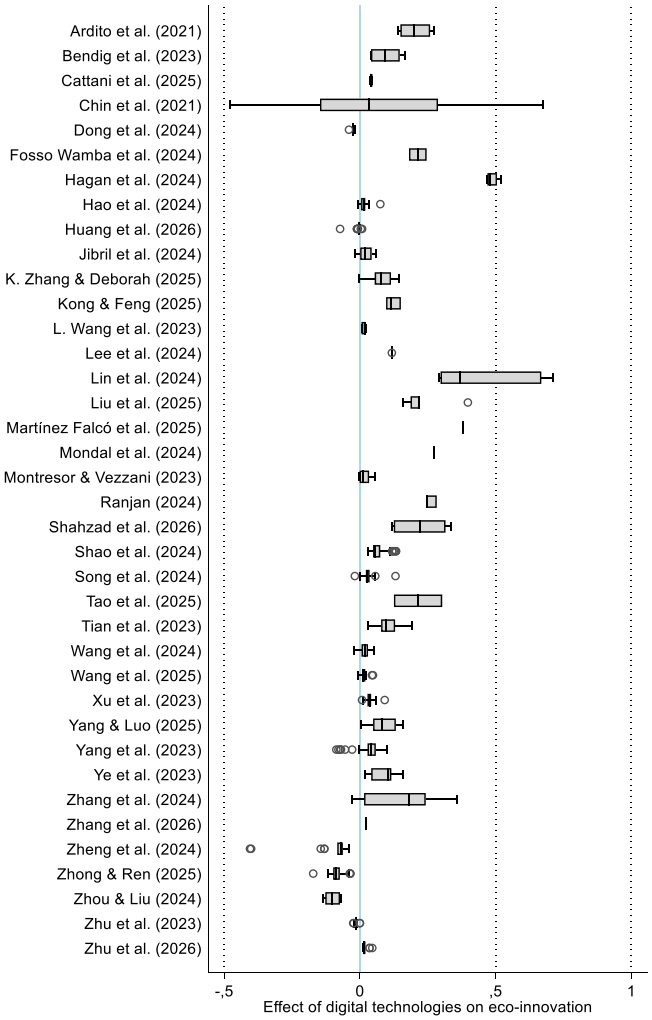


FIGURE 2 | Forest plot of digitalization effects on eco-innovation. Forest plot of the partial correlation coefficients estimated in the studies included in the meta-analysis on the effect of digital technologies on eco-innovation. Whiskers indicate the smallest and largest estimates. Circles indicate outliers.

eco-innovation, according to the individual studies included in this analysis (authors and years).

In general, most estimates fall to the right of zero, indicating a predominantly positive relationship between digital technologies and eco-innovation. Some studies identify small or nearly null effects; others indicate large effects, which might be due to methodological or contextual differences related to the country in which the sample was taken, the size or age of the company, or the type of digital technology or eco-innovation being studied.

The classic method for identifying publication bias relies on a funnel plot, in which each point represents an observation within a study (or the study itself when it reports a single estimate). The horizontal axis indicates effect magnitude, while the vertical axis reflects precision, measured in our case as the inverse of the standard error. More precise estimates appear higher and closer to the center, whereas less precise estimates

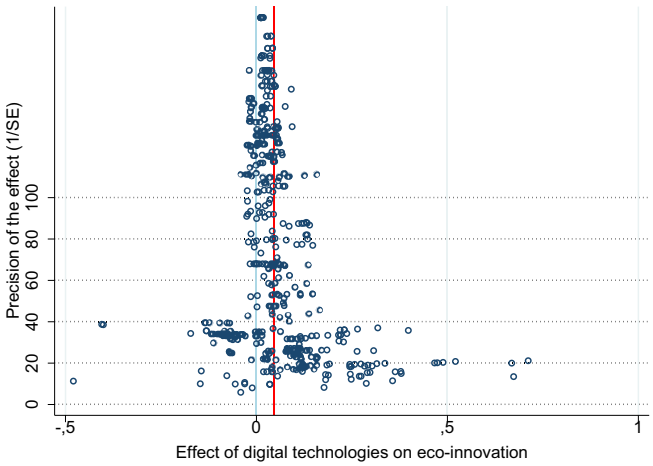


FIGURE 3 | Funnel plot of digitalization effects on eco-innovation. The x-axis represents the estimated effect of digitalization on eco-innovation across individual studies. The y-axis represents the precision of the effect, measured as the inverse of the standard error (1/SE), where higher values indicate greater precision.

are lower and more dispersed. A symmetrical funnel suggests no publication bias, whereas asymmetry indicates potential bias. In Figure 3, points are scattered toward the lower right, suggesting that less significant results may be underrepresented and, therefore, potential publication bias among the studies included in our analysis.

The statistical validity of our meta-analysis is strengthened by the inclusion of 38 studies, which includes numerous effect sizes (587). However, to avoid the subjectivity associated with a visual inspection of the graph, statistical tests (Egger et al. 1997) are useful for identifying asymmetry and calculating the magnitude of bias, using the equation (3):

$$r_i = \alpha_0 + \alpha_i \mathrm{SE}_i + \varepsilon_i \quad (3)$$

For this purpose, we test two hypotheses: the null hypothesis that $\alpha_1 = 0$ (which represents the FAT) and another null hypothesis that $\alpha_0 = 0$ (which represents the PET). The PET uses a regression similar to Equation (3) ($pcc_i = \beta_0 + \beta_1 SE_{pcc,i} + \varepsilon_i$) but changes the explanatory variable to the inverse of the standard error. If the null hypothesis (h_0) linked to any effect is not rejected in the PET (i.e., α_0 is nonsignificant and close to 0), then α_0 provides the estimate of the global effect. That is, if α_1 is significantly different from 0, publication bias likely exists. If α_0 is significantly different from 0, it offers evidence of a true effect, even if some bias remains. The FAT-PET could be sufficient if the SE coefficient (α_1) was very close to 0 (indicating an absence of bias) or if the constant (α_0) was significantly different from 0 (real effect). However, if the PET rejects null hypothesis (h_0) (i.e., α_0 is significantly different from 0), then we would use γ_0, as the best estimate of the global effect, which also is known as precision effect estimate with standard error (PEESE $pcc_i = \beta_0 + \beta_1 SE^2_{pcc,i} + \varepsilon_i$) (Acebo et al. 2021b; Stanley and Doucouliagos 2014). Rather than the standard error of PEESE, we use the variance of the estimated coefficient (SE_i^2), which provides a better estimate of the magnitude of

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TABLE 3 | Evidence of publication bias.

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Publication bias	1.535**	−0.285	1.841***	1.900**	1.829***		1.900***
(Standard error)	(0.605)	(0.506)	(0.675)	(0.830)	(0.672)		(0.251)
	[0.349 to 2.722]	[−1.311 to 0.740]	[0.519 to 3.163]	[0.274 to 3.526]	[0.512 to 3.147]		
Effect beyond bias	0.012	0.052***	0.006	0.005	0.006	0.007***	0.005***
(Standard error)	(0.011)	(0.011)	(0.006)	(0.006)	(0.006)	(0.002)	(0.001)
	[−0.008 to 0.033]	[0.030 to 0.075]	[−0.006 to 0.017]	[−0.007 to 0.018]	[−0.005 to 0.017]		
Observations	587	587	587	587	587	49	587
Studies	38	38	38	38	38	3	38

Note: This table reports results from different meta-regression approaches used to detect publication bias and estimate the underlying effect. OLS (ordinary least squares) refers to a standard regression based on the FAT-PET framework, where reported effect sizes are regressed on their standard errors. FE (fixed effects) controls for unobserved heterogeneity across studies by including study-level fixed effects. IV (instrumental variables) addresses potential endogeneity by instrumenting the reported standard errors with the inverse of the square root of sample size, a common proxy for statistical precision. Study-weighted (PEESE, pweight = weight) gives less influence to studies that report many estimates, weighting each observation by the inverse of the number of estimates per study and by their inverse variance. Precision-weighted (FAT-adjusted, pweight = precision_w) assigns more weight to more precise estimates (i.e., those with smaller standard errors), improving efficiency. In Panel B, WAAP (weighted average of adequately powered studies) reports the average effect using only studies with sufficient statistical power, reducing bias from imprecise estimates. Kink refers to the endogenous kink model, which allows the relationship between effect size and precision to change depending on the level of precision, capturing potential nonlinearities in publication bias.
Statistical significance: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

the effect, corrected for publication bias (Acebo et al. 2021b), as in Equation (4):

$$r_i = \gamma_0 + \gamma_i SE_i^2 + v_i \quad (4)$$

In Figure A1, we present a histogram of the estimated effects of digitalization on eco-innovation, displaying the significant heterogeneity, which is both asymmetric and positively skewed. Most estimates fall between 0 and 0.5, but some reach higher values. This asymmetry indicates the coexistence of studies reporting strong effects and others showing moderate or even negative impacts, possibly due to differences in sectors, company sizes, countries, or methodological approaches.

Table 3 reports the results of both linear and nonlinear regressions. In Panel A, the linear regressions include FAT-PET (with robust errors and clustering by study), FE (within studies, to control for unobservable heterogeneity between studies), IV (with corrections for heteroscedasticity and clustering), PEESE (with robust errors and clustering) regressions, as well as an OLS-type estimation with adjusted scale. The OLS, IV, PEESE, and adjusted scale test (Precision) indicate the strong publication bias. The mean true effect does not differ much from 0, and the FEE shows that bias is still present, even though the mean true effect appears positive and significant ($\beta_0 = 0.052$). Panel B contains the nonlinear regressions. The WAAP test shows a weak overall mean effect (0.007). The EK method suggests a positive but practically null true effect ($\beta_0 = 0.005$) and a strong, significant estimated publication bias ($\beta_1 = 1.900$). That is, the results available in extant literature appear inflated by publication bias.

4.3 | Heterogeneity Analysis

Rather than considering the reported estimates as a homogeneous set, the MRA allows us to examine whether the digitalization-eco-innovation association varies systematically across theoretically relevant subsets of the literature. Given the variability in prior findings, we examine heterogeneity across digital technologies, eco-innovation outcomes, econometric approaches, and control-variable specifications. Beyond identifying differences in estimated associations, we interpret these patterns in terms of the organizational and technological mechanisms through which digital resources may be translated into eco-innovation.

Table A7 reports the publication-bias tests for different technology subsets. The results reveal substantial heterogeneity across digital technologies. AI, digital innovation, industrial robotics, and smart product platforms generally show positive effects beyond publication bias in several specifications, although the evidence is less consistent for Industry 4.0 technologies and other digital orientations. Digital innovation displays positive and significant effects beyond bias across the linear specifications, while industrial robotics produces positive and significant effects in most models. Smart product platforms show particularly consistent positive effects across the linear specifications, with estimates ranging from 0.126 to 0.145. For Industry 4.0 technologies, evidence is more limited, with a significant positive effect beyond bias only in the FE specification ($\beta_0 = 0.581$). Other digital orientations also show more mixed results. The nonlinear WAAP estimates are nevertheless positive and statistically significant across

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the technological subgroups, with smart product platforms showing the largest adjusted estimate. EK estimates (Bom and Rachinger 2019) similarly indicate positive effects beyond publication bias for several categories, particularly AI, digital innovation, industrial robotics, and smart product platforms.

These differences can be understood from the nature and proximity of the technologies to the activities through which eco-innovation is generated. AI can support eco-innovation by analyzing large volumes of information, optimizing processes, predicting failures, reducing waste, and supporting the design of more sustainable products and processes. However, its effects need not be homogeneous because AI can also be deployed primarily for commercial or administrative purposes without producing eco-innovative outcomes. Similarly, digital innovation technologies and practices such as business process digitalization, CRM systems, servitization, digitization, e-commerce, informatization technology, Internet business models, and related applications intervene directly in business processes, customer relationships, distribution, demand management, and business models. Their greater proximity to organizational and operational activities may help explain the relatively consistent positive associations observed beyond publication bias.

Industrial robotics may display an even stronger association because it directly affects physical production processes. Automation and precision can reduce errors, defects, and material waste, thereby creating a relatively direct pathway from digital technology to environmental improvements. This pattern, however, should not be interpreted as causal evidence that robotics generates eco-innovation in all firms, since not all corrected estimates are positive and significant. By contrast, Industry 4.0 technologies (such as Big Data, Blockchain, Cloud Computing, CAD, the Internet, and Machine Learning) are largely general-purpose enabling technologies. They may be used for marketing, sales, information management, transactions, or other objectives that are not necessarily environmental. Their contribution to eco-innovation is therefore more dependent on how they are deployed within the firm, which may explain their more heterogeneous results.

A similar distinction applies to other digital orientations (such as digital orientation, digital investment, and digital government, among others). These variables represent conditions, resources, orientations, or capabilities rather than specific technological applications. Digital investment, for example, may support cybersecurity or marketing, while digital orientation may be directed towards commercial or administrative objectives. Likewise, technological innovation efficiency does not necessarily imply that the resulting innovations are environmental. These factors may therefore contribute indirectly to a firm's innovation capacity, while the greater distance from the specific outcome of eco-innovation makes their association weaker or more heterogeneous.

This pattern is consistent with both the RBV and DCV. From an RBV perspective, digital technologies are resources whose value depends on the complementary organizational, technological, and strategic resources with which they are combined. From a DCV perspective, the presence of digital resources is insufficient by itself because firms must be able to integrate, adapt, and reconfigure those resources to address changing conditions and generate innovation outcomes. Technologies that are more directly connected to production, products, or market-facing activities may therefore provide a shorter pathway to eco-innovation, whereas general-purpose technologies and broader digital orientations require complementary capabilities to be translated into environmental outcomes. The heterogeneity observed in Table A7 thus reflects not simply differences in technological labels, but differences in the distance between digital resources and the environmental outcome to which they are applied.

Table A8 reveals a second important source of heterogeneity, as the association differs according to the type of eco-innovation outcome considered. For green innovation, the effect beyond publication bias is positive and statistically significant in four of the five linear specifications, while the IV estimate is positive but not significant. This pattern is also supported by the WAAP estimate (β_0= 0.014) and the EK estimate (α_1=0.012). For decarbonization, however, the evidence is substantially weaker and less consistent, with negative and statistically significant estimates in some specifications, including both WAAP (β_0 = −0.003) and EK (α_1 = −0.003). The EK results further indicate significant publication bias for green innovation (δ = 2.562), whereas the corresponding coefficient for decarbonization is not significant (δ = −0.086).

This difference can be interpreted in terms of the distance between digitalization and the environmental outcome. Digitalization may be more readily incorporated into business processes, products, and innovation activities that support green innovation, whereas decarbonization represents a more specific environmental outcome whose achievement may require a closer alignment between digital capabilities and the reduction of carbon emissions. Thus, digitalization may be associated with the development of environmentally oriented innovations without necessarily producing a comparable association with direct emissions reductions. The contrasting WAAP and EK estimates provide additional evidence that this distinction persists after accounting for publication bias. From an RBV perspective, the difference reflects the fact that the value of digital resources depends on how they are deployed towards specific outcomes; from a DCV perspective, achieving decarbonization may require firms to reconfigure and align their resources more specifically with environmental and operational objectives (Fosso Wamba, Queiroz, and Trinchera 2024).

The variation across econometric approaches provides an additional indication of heterogeneity. As shown in Table A9, the estimated effects differ across OLS, FE, IV, Probit/Logit, and PLS-SEM approaches. OLS and PLS-SEM specifications generally report positive and significant effects beyond publication bias, whereas FE, IV, and Probit/Logit results are more mixed. This variation should not be interpreted as evidence that one estimation method necessarily provides the “true” effect. Rather, it indicates that the estimated digitalization-eco-innovation association varies across empirical settings and modelling approaches.

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This methodological heterogeneity is also consistent with the RBV and DCV perspectives, which is reflected in differences in complementary resources and in firms' ability to integrate, adapt, and reconfigure digital resources, which may contribute to the variation observed across empirical settings. Thus, firms with apparently similar levels of digitalization may not obtain similar eco-innovation outcomes.

Finally, Table A10 shows that heterogeneity is also related to the control variables included in the primary studies. Evidence of publication bias is more frequently observed among studies controlling for external collaboration, exporting, ownership type, age, and firm size, particularly in OLS specifications. More importantly, the effects beyond publication bias show a relatively consistent positive pattern for R&D, environmental regulation, ownership type, and firm size across OLS and FE specifications, while firm age is positive and significant in the FE estimates. These results suggest that the digitalization-eco-innovation association is embedded in firm-level and contextual conditions rather than operating in isolation. This pattern reinforces the RBV and DCV interpretation developed above. From an RBV perspective, complementary resources such as R&D capabilities, external financing, and even firm size may increase the value that firms derive from digital technologies. From a DCV perspective, firms must be able to mobilize and reconfigure these resources in response to technological, organizational, and environmental conditions; this is reflected in the observed results, for instance, regarding environmental regulations. Accordingly, digitalization should not be understood as an isolated driver of eco-innovation. Its contribution depends on the resources with which it is combined, the capabilities through which those resources are deployed, and the specific environmental objective pursued.

In aggregate, the heterogeneity analysis indicates that the digitalization-eco-innovation association is positive but not uniform. Differences emerge according to the type of digital technology, the environmental outcome, the econometric approach, and the complementary conditions represented by control variables. The results are therefore consistent with a view in which digitalization provides resources and enabling capabilities, but their contribution to eco-innovation depends on their proximity to environmental activities and, crucially, on firms' ability to integrate, deploy, and reconfigure them. This interpretation provides a theoretical explanation for why digitalization may be positively associated with eco-innovation overall while producing substantially different associations across technologies, outcomes, and organizational contexts.

5 | Discussion

Our investigation examines whether the association between digitalization and eco-innovation persists after accounting for publication bias and how it varies across digital technologies and environmental outcomes. The overall unweighted mean PCC is positive and statistically significant (0.046). However, after weighting estimates by precision using fixed-effects and weighted least squares approaches, the average association decreases substantially (0.008), although it remains positive and statistically significant. This indicates a positive but relatively small association between digitalization and eco-innovation.

For the full sample of 38 studies (587 effect estimates), evidence from both linear and nonlinear models indicates publication bias in several specifications (Table 3). Nevertheless, the FE estimator and the nonlinear WAAP and Kink tests show that positive associations remain after accounting for publication bias, supporting the robustness of the overall positive association, although its magnitude is relatively small.

Publication bias may arise in this literature because studies reporting positive or statistically significant associations between digitalization and eco-innovation may be more likely to be submitted, published, and subsequently identified than studies reporting weak, null, or negative associations. This tendency may be particularly relevant in a rapidly developing field such as the twin transition, where digitalization is often expected to support environmental innovation and positive findings may therefore attract greater academic and practical interest. The presence of publication bias identified in Table 3 suggests that the observed positive association in the literature should consequently be interpreted with caution, reinforcing the importance of correcting for selective reporting when assessing the underlying evidence.

The results also indicate heterogeneity across digital technologies (Table A7 in appendix). For AI, the positive association after accounting for publication bias is consistent with Montresor and Vezzani (2023) findings of a positive but heterogeneous relationship with eco-innovation and with Lin et al.'s (2024) and Jibril et al.'s (2024) findings regarding green innovation. Industrial robotics shows a positive but heterogeneous association, partially contrasting with Huang et al. (2026) but more consistent with L. Wang et al. (2023). Smart product platforms exhibit the largest and most consistent estimates across the reported specifications; however, this result should be interpreted cautiously because it is based on a single primary study and therefore does not provide stronger evidence of robustness than other technology categories.

With regard to environmental outcomes, the association is more consistently observed for green innovation, as shown in Table A8 and in line with Ardito et al. (2021), Hao et al. (2024), and Ye et al. (2023). By contrast, evidence for decarbonization is weaker and less consistent, with small and often nonsignificant negative estimates. This pattern is consistent with studies reporting negative associations between digitalization and carbon emissions (G. Yang et al. 2023; D. Zheng et al. 2024; Zhou and Liu 2024; Q. Zhu et al. 2023). Overall, the findings suggest that digitalization is positively associated with eco-innovation, but that the magnitude and direction of this association depend on the technology and environmental outcome considered.

6 | Conclusions

This study offers a quantitative synthesis of recent literature on the relationship between digitalization and eco-innovation, based on 587 effect sizes reported by 38 studies published in

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high-impact journals. Our MRA provides evidence of publication bias; after accounting for this bias, the results indicate a small, positive, and statistically significant association between digitalization and eco-innovation.

The results also reveal substantial heterogeneity across technologies and environmental outcomes. Digital innovation shows the most consistent positive association after accounting for publication bias, with positive and statistically significant estimates across all reported specifications. AI and industrial robotics also show positive associations, although their estimates vary across specifications. Industry 4.0 technologies show weaker and less consistent results. Smart product platforms display the largest estimates across the reported specifications; however, these results are based on a single primary study and therefore should not be interpreted as evidence of greater robustness than that for other technology categories.

The distinction between green innovation and decarbonization further qualifies the overall positive association. Digitalization is more consistently associated with green innovation, whereas estimates for decarbonization are smaller, negative, and less consistent. For example, the estimates beyond publication bias for green innovation are predominantly positive and statistically significant, ranging from 0.012 to 0.088 across the reported specifications, whereas decarbonization shows negative and statistically significant estimates in several of the models. This pattern indicates that digitalization is more consistently associated with green innovation than with decarbonization. Thus, statistical significance should not be equated with substantial practical or managerial significance, as the estimated overall association is relatively small, and the results do not indicate that digitalization alone is sufficient to achieve environmental improvements or emissions reductions.

From a theoretical perspective, the findings provide a more qualified interpretation of the twin transition through the RBV and DCV. The heterogeneity across technologies and environmental outcomes suggests that digital resources do not have a uniform association with eco-innovation. For example, estimates beyond publication bias for digital innovation are positive and statistically significant across all reported specifications (ranging from 0.014 to 0.031), whereas smart product platforms show larger estimates (ranging from 0.126 to 0.145), but are based on a single primary study (Table A7 in appendix). Consistent with the RBV, their contribution depends on the complementary organizational and technological resources with which they are combined, as digitalization is associated with green technology innovation through improved resource management (Shao et al. 2024). From a DCV perspective, the results further suggest that firms' ability to integrate, deploy, adapt, and reconfigure digital resources is relevant to how these resources are associated with eco-innovation (Fosso Wamba, Queiroz, and Trinchera 2024).

From a managerial perspective, the findings suggest that statistical significance alone should not determine technology investment priorities. The relatively small overall association and the heterogeneity across technologies indicate that managers should consider how digital resources are integrated and deployed within specific organizational and environmental contexts, including employee digital training, organizational capabilities, and environmental investment. The stronger evidence for green innovation, together with the weaker evidence for decarbonization, also suggests that digitalization should not be regarded as sufficient by itself to achieve emissions reductions, but rather as part of broader complementary efforts, such as investment in environmental improvements, workforce training, and organizational changes that support the effective use of digital capabilities.

Future studies should further examine the association between digitalization and eco-innovation by considering a broader range of empirical contexts and research designs. In particular, longitudinal, cross-sectional, and survey-based evidence could be complemented by additional research designs to provide a more comprehensive understanding of how these associations vary across different settings. Further research should also expand the geographical scope of the evidence, particularly in underrepresented regions such as Latin America, to assess the extent to which existing findings hold across different institutional and economic contexts. In addition, future meta-analyses could consider studies published in languages other than English to broaden the evidence base and reduce the potential influence of language restrictions.

Author Contributions

Rafael Alberto Iturralde Solorzano: investigation, writing – original draft, data curation, methodology, conceptualization. **Enrique Acebo:** writing – review and editing, funding acquisition, conceptualization, methodology, supervision. **José-Ángel Miguel-Dávila:** conceptualization, funding acquisition, methodology, writing – review and editing, supervision.

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Disclosure

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Ethics Statement

This meta-analysis has been conducted in accordance with the reporting guidelines established by the journal. Particular attention has been given to methodological transparency, the detailed description of inclusion and exclusion criteria, the procedures for data collection and

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analysis, as well as the presentation and interpretation of results. The present study was not prospectively registered in an international database; however, all procedures followed established guidelines for systematic reviews and meta-analyses (PRISMA 2020). All methodological decisions, including search strategies, eligibility criteria, and statistical approaches, were established prior to data collection and applied consistently throughout the study.

In the development of this study, tools based on AI have been used occasionally, solely for the purpose of supporting the structuring and coherence of the text, and under no circumstances have they been used for the generation of substantive content.

Data Availability Statement

The dataset supporting the findings of this study is provided as Supporting Information submitted to the journal. It was constructed from published studies included in the meta-analysis, following the procedures described in the manuscript.

References

Acebo, E., J. Á. Miguel-Dávila, and M. Nieto. 2021a. "External Stakeholder Engagement: Complementary and Substitutive Effects on Firms' Eco-Innovation." *Business Strategy and the Environment* 30, no. 5: 2671–2687. https://doi.org/10.1002/bse.2770.

Acebo, E., J. Á. Miguel-Dávila, and M. Nieto. 2021b. "The Impact of University-Industry Relationships on Firms' Performance: A Meta-Regression Analysis." *Science and Public Policy* 48, no. 2: 276–293. https://doi.org/10.1093/scipol/scab025.

Appio, F. P., F. Frattini, A. M. Petruzzelli, and P. Neirotti. 2021. "Digital Transformation and Innovation Management: A Synthesis of Existing Research and an Agenda for Future Studies." *Journal of Product Innovation Management* 38, no. 1: 4–20). Blackwell Publishing Ltd. https://doi.org/10.1111/jpim.12562.

Ardito, L., S. Raby, V. Albino, and B. Bertoldi. 2021. "The Duality of Digital and Environmental Orientations in the Context of SMEs: Implications for Innovation Performance." *Journal of Business Research* 123: 44–56. https://doi.org/10.1016/j.jbusres.2020.09.022.

Barney, J. 1991. "Firm Resources and Sustained Competitive Advantage." *Journal of Management* 17, no. 1: 99–120.

Bendig, D., C. Schulz, L. Theis, and S. Raff. 2023. "Digital Orientation and Environmental Performance in Times of Technological Change." *Technological Forecasting and Social Change* 188: 122272. https://doi.org/10.1016/j.techfore.2022.122272.

Bom, P. R. D., and H. Rachinger. 2019. "A Kinked Meta-Regression Model for Publication Bias Correction." *Research Synthesis Methods* 10, no. 4: 497–514. https://doi.org/10.1002/jrsm.1352.

Cattani, L., S. Montresor, and A. Vezzani. 2025. "Firms' Eco-Innovation and Industry 4.0 Technologies in Urban and Rural Areas." *Regional Studies* 59, no. 1. https://doi.org/10.1080/00343404.2023.2243984.

Chen, D. 2022. "Big Data Analysis on the Effect of Cost Stickiness on Digital Transformation." *Mobile Information Systems* 2022: 1–11. https://doi.org/10.1155/2022/5883315.

Chin, T., W. Wang, M. Yang, Y. Duan, and Y. Chen. 2021. "The Moderating Effect of Managerial Discretion on Blockchain Technology and the Firms' Innovation Quality: Evidence From Chinese Manufacturing Firms." *International Journal of Production Economics* 240: 108219. https://doi.org/10.1016/j.ijpe.2021.108219.

D'Agostino, L. M., and R. Moreno. 2019. "Green Regions and Local Firms' Innovation." *Papers in Regional Science* 98, no. 4: 1585–1608. https://doi.org/10.1111/pirs.12427.

Demirel, P., E. Martinez-Ros, and F. Quatraro. 2025. "Innovation for the Green Transition: Challenges and Future Perspectives." *Eurasian Business Review* 15: 631–645. https://doi.org/10.1007/s40821-025-00322-w.

Di Vaio, A., R. Palladino, R. Hassan, and O. Escobar. 2020. "Artificial Intelligence and Business Models in the Sustainable Development Goals Perspective: A Systematic Literature Review." *Journal of Business Research* 121: 283–314. https://doi.org/10.1016/j.jbusres.2020.08.019.

Ding, N., P. Guo, Y. H. Xi, A. Zhang, and X. Lei. 2024. "Low-Carbon Development in Power Systems Based on Carbon Emission Analysis Models: A Comprehensive Review." In *Sustainable Energy Technologies and Assessments*, vol. 65. Elsevier Ltd. https://doi.org/10.1016/j.seta.2024.103774.

Dong, J., J. Tian, H. Lin, X. Suo, H. Wu, and R. Zeng. 2024. "Can Corporate Digital Innovation Speed Up Climate Change Mitigation? Evidence From China." *IEEE Transactions on Engineering Management* 71: 7797–7814. https://doi.org/10.1109/TEM.2024.3386181.

Egger, M., G. Davey Smith, C. E. Minder, M. Schneider, and C. Minder. 1997. "Bias in Meta-Analysis Detected by a Simple, Graphical Test." *BMJ* 315, no. 7109: 629–634. https://doi.org/10.1136/bmj.315.7109.629.

European Commission. 2019. "The European Green Deal." https://commission.europa.eu/publications/communication-european-green-deal_en.

Fichman, R. G., B. L. Dos Santos, and Z. Zheng. 2014. "Digital Innovation as a Fundamental and Powerful Concept in the Information Systems Curriculum." *MIS Quarterly: Management Information Systems* 38, no. 2: 329–353). University of Minnesota. https://doi.org/10.25300/misq/2014/38.2.01.

Filonchyk, M., M. P. Peterson, L. Zhang, V. Hurynovich, and Y. He. 2024. "Greenhouse Gases Emissions and Global Climate Change: Examining the Influence of CO2, CH4, and N2O." In *Science of the Total Environment*, vol. 935. Elsevier B.V. https://doi.org/10.1016/j.scitotenv.2024.173359.

Fosso Wamba, S., M. M. Queiroz, I. O. Pappas, and Y. Sullivan. 2024. "Artificial Intelligence Capability and Firm Performance: A Sustainable Development Perspective by the Mediating Role of Data-Driven Culture." *Information Systems Frontiers* 26: 2189–2203. https://doi.org/10.1007/s10796-023-10460-z.

Fosso Wamba, S., M. M. Queiroz, and L. Trinchera. 2020. "Dynamics Between Blockchain Adoption Determinants and Supply Chain Performance: An Empirical Investigation." *International Journal of Production Economics* 229: 107791. https://doi.org/10.1016/j.ijpe.2020.107791.

Fosso Wamba, S., M. M. Queiroz, and L. Trinchera. 2024. "The Role of Artificial Intelligence-Enabled Dynamic Capability on Environmental Performance: The Mediation Effect of a Data-Driven Culture in France and the USA." *International Journal of Production Economics* 268: 109131. https://doi.org/10.1016/j.ijpe.2023.109131.

Gao, P., J. He, M. Vochozka, and S. Hu. 2024. "Deterrent Effects of Central Environmental Protection Inspection on Green Technology Innovation Cycles: Evidence From the Listed State-Owned Firms in China." *Technological Forecasting and Social Change* 209: 123783. https://doi.org/10.1016/j.techfore.2024.123783.

García-Granero, E. M., L. Piedra-Muñoz, and E. Galdeano-Gómez. 2020. "Measuring Eco-Innovation Dimensions: The Role of Environmental Corporate Culture and Commercial Orientation." *Research Policy* 49, no. 8: 104028. https://doi.org/10.1016/j.respol.2020.104028.

Giustiziero, G., T. Kretschmer, D. Somaya, and B. Wu. 2023. "Hyperspecialization and Hyperscaling: A Resource-Based Theory of the Digital Firm." *Strategic Management Journal* 44, no. 6: 1391–1424. https://doi.org/10.1002/smj.3365.

10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Hagan, M. C., F. K. Andoh-Baidoo, E. W. Ayaburi, and J. K. Nwankpa. 2024. "A Moderated Model of Open Innovation: An Integration of Theories of Constraint and Dynamic Capabilities." *IEEE Transactions on Engineering Management* 71: 14438–14449. https://doi.org/10.1109/TEM.2024.3440070.

Haleem, F., M. Ilyas, and M. Jehangir. 2024. "Driving Organizational Value Through Sustainable Business Processes and Green Information Technology." *Business Process Management Journal* 30, no. 7: 2380–2400. https://doi.org/10.1108/BPMJ-01-2024-0011.

Hameed, Z., R. M. Naeem, P. Mishra, V. Chotia, and A. Malibari. 2023. "Ethical Leadership and Environmental Performance: The Role of Green IT Capital, Green Technology Innovation, and Technological Orientation." *Technological Forecasting and Social Change* 194: 122739. https://doi.org/10.1016/j.techfore.2023.122739.

Hao, X., E. Miao, Q. Sun, K. Li, S. Wen, and Y. Xue. 2024. "The Impact of Digital Government on Corporate Green Innovation: Evidence From China." *Technological Forecasting and Social Change* 206: 123570. https://doi.org/10.1016/j.techfore.2024.123570.

Havránek, T., T. D. Stanley, H. Doucouliagos, et al. 2020. "Reporting Guidelines for Meta-Analysis in Economics." *Journal of Economic Surveys* 34, no. 3: 469–475. https://doi.org/10.1111/joes.12363.

Horbach, J. 2024. "Digitalisation and Sustainability Measures at the Firm Level." *Digital Economy and Sustainable Development* 2, no. 1: 19. https://doi.org/10.1007/s44265-024-00043-4.

Huang, R., Q. Miao, and X. Yao. 2026. "Cutting Emissions Through Intelligent Production in Chinese Manufacturing Firms: An Empirical Analysis." *Annals of Operations Research* 359: 1849–1872. https://doi.org/10.1007/s10479-024-06005-5.

Hund, A., H. T. Wagner, D. Beimborn, and T. Weitzel. 2021. "Digital Innovation: Review and Novel Perspective." *Journal of Strategic Information Systems* 30, no. 4: 101695. https://doi.org/10.1016/j.jsis.2021.101695.

Ionaşcu, M., I. Ionaşcu, E. Turuianu, and M. Săcărin. 2026. "Digital Transformation and Eco-Innovation: Evidence on the Twin Transition of European Companies." *Corporate Social Responsibility and Environmental Management*: e71003. https://doi.org/10.1002/csr.71003.

Irsova, Z., H. Doucouliagos, T. Havranek, and T. D. Stanley. 2024. "Meta-Analysis of Social Science Research: A Practitioner's Guide." *Journal of Economic Surveys* 38, no. 5: 1547–1566. https://doi.org/10.1111/joes.12595.

Jibril, H., E. Kesidou, and S. Roper. 2024. "Do Digital Technologies Enable Firms That Prioritize Sustainability Goals to Innovate? Empirical Evidence From Established UK Micro-businesses." *British Journal of Management* 35, no. 4: 2155–2173. https://doi.org/10.1111/1467-8551.12821.

Kemp, R., and P. Pearson. 2007. "Final Report MEI Project About Measuring Eco-Innovation." Project Deliverable. http://www.oecd.org/env/consumption-innovation/43960830.pdf.

Kong, T., and T. Feng. 2025. "Linking Digital Transformation With Supply Chain Intelligence Integration and Carbon Performance: Does Business Model Innovation Matter?" *Business Process Management Journal* 31, no. 4: 1274–1305. https://doi.org/10.1108/BPMJ-09-2023-0682.

Lee, C. C., S. Qin, and Y. Li. 2022. "Does Industrial Robot Application Promote Green Technology Innovation in the Manufacturing Industry?" *Technological Forecasting and Social Change* 183: 121893. https://doi.org/10.1016/j.techfore.2022.121893.

Lee, M. J., Y. Kim, and T. Roh. 2024. "Exploring the Role of Digital Servitization for Green Innovation: Absorptive Capacity, Transformative Capacity, and Environmental Strategy." *Technological Forecasting and Social Change* 207: 123614. https://doi.org/10.1016/j.techfore.2024.123614.

Li, H., Y. Su, C. J. Ding, G. G. Tian, and Z. Wu. 2024. "Unveiling the Green Innovation Paradox: Exploring the Impact of Carbon Emission Reduction on Corporate Green Technology Innovation." *Technological Forecasting and Social Change* 207: 123562. https://doi.org/10.1016/j.techfore.2024.123562.

Lin, B., and Y. Xie. 2024. "Driving Green Technology Innovation in Renewable Energy: Does Venture Capital Matter?" *IEEE Transactions on Engineering Management* 71: 6784–6795. https://doi.org/10.1109/TEM.2023.3286978.

Lin, J., Y. Zeng, S. Wu, and X. Luo. 2024. "How Does Artificial Intelligence Affect the Environmental Performance of Organizations? The Role of Green Innovation and Green Culture." *Information & Management* 61, no. 2: 103924. https://doi.org/10.1016/j.im.2024.103924.

Liu, J., H. Chang, J. Y. L. Forrest, and B. Yang. 2020. "Influence of Artificial Intelligence on Technological Innovation: Evidence From the Panel Data of China's Manufacturing Sectors." *Technological Forecasting and Social Change* 158: 120142. https://doi.org/10.1016/j.techfore.2020.120142.

Liu, Y., X. Zhang, and Y. Shen. 2024. "Technology-Driven Carbon Reduction: Analyzing the Impact of Digital Technology on China's Carbon Emission and Its Mechanism." *Technological Forecasting and Social Change* 200: 123124. https://doi.org/10.1016/j.techfore.2023.123124.

Liu, Z., S. Han, M. Yao, S. Gupta, and I. Laguir. 2025. "Exploring Drivers of Eco-Innovation in Manufacturing Firms' Circular Economy Transition: An Awareness, Motivation, Capability Perspective." *Annals of Operations Research* 355: 147–182. https://doi.org/10.1007/s10479-023-05473-5.

Lubacha, J., and T. Wendler. 2021. "Do European Firms Obey the Rules? Environmental Innovativeness in Light of Institutional Frameworks." *Industry and Innovation* 28, no. 9: 1196–1223. https://doi.org/10.1080/13662716.2021.1929869.

Martínez Falcó, J., E. Sánchez-García, B. Marco-Lajara, and U. Akram. 2025. "Digital Transformation and Green Innovation Performance: Unraveling the Role of Green Knowledge Sharing and Top Management Environmental Awareness." *Internet Research* 35, no. 4: 1621–1656. https://doi.org/10.1108/INTR-11-2023-1016.

Mikalef, P., and M. Gupta. 2021. "Artificial Intelligence Capability: Conceptualization, Measurement Calibration, and Empirical Study on Its Impact on Organizational Creativity and Firm Performance." *Information and Management* 58, no. 3: 103434. https://doi.org/10.1016/j.im.2021.103434.

Mondal, S., S. Singh, and H. Gupta. 2024. "Achieving Technological Transformation and Social Sustainability: An Industry 4.0 Perspective." *IEEE Transactions on Engineering Management* 71: 6623–6635. https://doi.org/10.1109/TEM.2023.3246637.

Montresor, S., and A. Vezzani. 2023. "Digital Technologies and Eco-Innovation. Evidence of the Twin Transition From Italian Firms." *Industry and Innovation* 30, no. 7: 766–800. https://doi.org/10.1080/13662716.2023.2213179.

Nasiri, M., M. Saunila, T. Rantala, and J. Ukko. 2022. "Sustainable Innovation Among Small Businesses: The Role of Digital Orientation, the External Environment, and Company Characteristics." *Sustainable Development* 30, no. 4: 703–712. https://doi.org/10.1002/sd.2267.

Ning, L., and R. Guo. 2022. "Technological Diversification to Green Domains: Technological Relatedness, Invention Impact and Knowledge Integration Capabilities." *Research Policy* 51, no. 1: 104406. https://doi.org/10.1016/j.respol.2021.104406.

Opatrny, M., T. Havranek, Z. Irsova, and M. Scasny. 2025. "Publication Bias and Model Uncertainty in Measuring the Effect of Class Size on Achievement." *Journal of Labor Economics*. https://doi.org/10.1086/737989.

10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Opazo-Basáez, M., J. C. Monroy-Osorio, and J. Marić. 2024. "Evaluating the Effect of Green Technological Innovations on Organizational and Environmental Performance: A Treble Innovation Approach." *Technovation* 129: 102885. https://doi.org/10.1016/j.technovation.2023.102885.

Page, M. J., J. E. McKenzie, P. M. Bossuyt, et al. 2021. "The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews." *Revista Española de Cardiología* 74, no. 9: 790–799. https://doi.org/10.1016/j.recesp.2021.06.016.

Ranjan, P. 2024. "Unraveling the Mystery of the Link Between Digital Orientation and Innovation Performance: The Interplay of Digital Business Capability and Environmental Dynamism." *Technovation* 131: 102966. https://doi.org/10.1016/j.technovation.2024.102966.

Shahzad, F., Y. Ben Zaied, and Q. Zhang. 2026. "Integrating Industry 4.0 Technologies Into Green Supply Chains: An Empirical Investigation of Pathways to Green Innovation and Circular Economy." *International Journal of Production Economics* 291: 109859. https://doi.org/10.1016/j.ijpe.2025.109859.

Shao, Y., K. Xu, and Y. G. Shan. 2024. "Leveraging Corporate Digitalization for Green Technology Innovation: The Mediating Role of Resource Endowments." *Technovation* 133: 102999. https://doi.org/10.1016/j.technovation.2024.102999.

Sharaf-Addin, H. H. 2024. "Towards Net-Zero Carbon Emissions: A Systematic Review of Carbon Sustainability Reporting Based on GHG Protocol Framework." In *Environmental and Sustainability Indicators*, vol. 24. Elsevier B.V. https://doi.org/10.1016/j.indic.2024.100516.

Sheng, J., R. Ding, and H. Yang. 2024. "Corporate Green Innovation in an Aging Population: Evidence From Chinese Listed Companies." *Technological Forecasting and Social Change* 202: 123307. https://doi.org/10.1016/j.techfore.2024.123307.

Siedschlag, I., G. Mohan, and W. Yan. 2024. "Twin Transitions Across Enterprises: Do Digital Technologies and Sustainability Go Together?" *Journal of Cleaner Production* 481: 144025. https://doi.org/10.1016/j.jclepro.2024.144025.

Škare, M., B. Gavurova, and M. Porada-Rochon. 2024. "Digitalization and Carbon Footprint: Building a Path to a Sustainable Economic Growth." *Technological Forecasting and Social Change* 199: 123045. https://doi.org/10.1016/j.techfore.2023.123045.

Song, Y., C. Du, P. Du, R. Liu, and Z. Lu. 2024. "Digital Transformation and Corporate Environmental Performance: Evidence From Chinese Listed Companies." *Technological Forecasting and Social Change* 201: 123159. https://doi.org/10.1016/j.techfore.2023.123159.

Stanley, T. D., and H. Doucouliagos. 2014. "Meta-Regression Approximations to Reduce Publication Selection Bias." *Research Synthesis Methods* 5, no. 1: 60–78. https://doi.org/10.1002/jrsm.1095.

Tao, J., W. Huang, L. Xu, and H. Ren. 2025. "Sensing and Seizing Opportunities for Green Innovation: The Role of Business Process Digitalization." *IEEE Transactions on Engineering Management* 72: 2798–2810. https://doi.org/10.1109/TEM.2025.3583782.

Teece, D. J., G. Pisano, and A. Shuen. 1997. "Dynamic Capabilities and Strategic Management." *Strategic Management Journal* 18, no. 7: 509–533. http://www.jstor.org/stable/3088148.

Tian, H., L. Zhao, L. Yunfang, and W. Wang. 2023. "Can Enterprise Green Technology Innovation Performance Achieve "Corner Overtaking" by Using Artificial Intelligence?—Evidence From Chinese Manufacturing Enterprises." *Technological Forecasting and Social Change* 194: 122732. https://doi.org/10.1016/j.techfore.2023.122732.

Tranfield, D., D. Denyer, and P. Smart. 2003. "Towards a Methodology for Developing Evidence-Informed Management Knowledge by Means of Systematic Review." *British Journal of Management* 14, no. 3: 207–222. https://doi.org/10.1111/1467-8551.00375.

Turkcan, H., S. Z. Imamoglu, and H. Ince. 2022. "To Be More Innovative and More Competitive in Dynamic Environments: The Role of Additive Manufacturing." *International Journal of Production Economics* 246: 108418. https://doi.org/10.1016/j.ijpe.2022.108418.

Verhoef, P. C., T. Broekhuizen, Y. Bart, et al. 2021. "Digital Transformation: A Multidisciplinary Reflection and Research Agenda." *Journal of Business Research* 122: 889–901. https://doi.org/10.1016/j.jbusres.2019.09.022.

Wang, B., S. Gong, and Y. Yang. 2024. "Unveiling the Relation Between Digital Technology and Low-Carbon Innovation: Carbon Emission Trading Policy as an Antecedent." *Technological Forecasting and Social Change* 205: 123522. https://doi.org/10.1016/j.techfore.2024.123522.

Wang, E. Z., C. C. Lee, and Y. Li. 2022. "Assessing the Impact of Industrial Robots on Manufacturing Energy Intensity in 38 Countries." *Energy Economics* 105: 105748. https://doi.org/10.1016/j.eneco.2021.105748.

Wang, J., H. Tian, and P. Zheng. 2025. "The Impact of Artificial Intelligence on Corporate Environmental Investment." *IEEE Transactions on Engineering Management* 72: 96–114. https://doi.org/10.1109/TEM.2024.3491940.

Wang, L., Q. Chen, Z. Dong, and L. Cheng. 2024. "The Role of Industrial Intelligence in Peaking Carbon Emissions in China." *Technological Forecasting and Social Change* 199: 123005. https://doi.org/10.1016/j.techfore.2023.123005.

Wang, L., and T. S. Ramsey. 2024. "Digital Divide and Environmental Pressure: A Countermeasure on the Embodied Carbon Emissions in FDI." *Technological Forecasting and Social Change* 204: 123398. https://doi.org/10.1016/j.techfore.2024.123398.

Wang, L., Y. Zhou, and B. Chiao. 2023. "Robots and Firm Innovation: Evidence From Chinese Manufacturing." *Journal of Business Research* 162: 113878. https://doi.org/10.1016/j.jbusres.2023.113878.

Wang, S., M. Chen, and M. Song. 2018. "Energy Constraints, Green Technological Progress and Business Profit Ratios: Evidence From Big Data of Chinese Enterprises." *International Journal of Production Research* 56, no. 8: 2963–2974. https://doi.org/10.1080/00207543.2018.1454613.

Xu, Y., L. Yuan, R. Khalfaoui, M. Radulescu, S. Mallek, and X. Zhao. 2023. "Making Technological Innovation Greener: Does Firm Digital Transformation Work?" *Technological Forecasting and Social Change* 197: 122928. https://doi.org/10.1016/j.techfore.2023.122928.

Yang, G., Y. Nie, H. Li, and H. Wang. 2023. "Digital Transformation and Low-Carbon Technology Innovation in Manufacturing Firms: The Mediating Role of Dynamic Capabilities." *International Journal of Production Economics* 263: 108969. https://doi.org/10.1016/j.ijpe.2023.108969.

Yang, H., L. Li, and Y. Liu. 2022. "The Effect of Manufacturing Intelligence on Green Innovation Performance in China." *Technological Forecasting and Social Change* 178: 121569. https://doi.org/10.1016/j.techfore.2022.121569.

Yang, Y., and F. Luo. 2025. "Bridging Digital Transformation and Green Innovation in Firms: A Dual-Dimensional Analysis of Disclosure-Behavior Alignment." *Journal of Technology Transfer* 51: 3278–3304. https://doi.org/10.1007/s10961-025-10278-8.

Ye, F., Y. Ouyang, and Y. Li. 2023. "Digital Investment and Environmental Performance: The Mediating Roles of Production Efficiency and Green Innovation." *International Journal of Production Economics* 259: 108822. https://doi.org/10.1016/j.ijpe.2023.108822.

Yuan, B., and X. Cao. 2022. "Do Corporate Social Responsibility Practices Contribute to Green Innovation? The Mediating Role of Green Dynamic Capability." *Technology in Society* 68: 101868. https://doi.org/10.1016/j.techsoc.2022.101868.

10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Zhang, H., Y. Lv, J. Z. Zhang, and P. Fadil. 2026. "Digital Government Development, Corporate Digital Transformation, and Carbon Performance: Insights From Listed Companies in China." *Technological Forecasting and Social Change* 223: 124414. https://doi.org/10.1016/j.techfore.2025.124414.

Zhang, K., and G. Deborah. 2025. "Digitalization, Internationalization and Green Innovation in China: An Analysis Based on Threshold and Mediation Effects." *Journal of Technology Transfer* 50: 962–994. https://doi.org/10.1007/s10961-024-10116-3.

Zhang, W., S. Ye, S. K. Mangla, A. Emrouznejad, and M. Song. 2024. "Smart Platforming in Automotive Manufacturing for NetZero: Intelligentization, Green Technology, and Innovation Dynamics." *International Journal of Production Economics* 274: 109289. https://doi.org/10.1016/j.ijpe.2024.109289.

Zheng, D., H. Song, C. Zhao, Y. Liu, and W. Zhao. 2024. "Is It Possible for Semiconductor Companies to Reduce Carbon Emissions Through Digital Transformation? Evidence From China." *International Journal of Production Economics* 272: 109246. https://doi.org/10.1016/j.ijpe.2024.109246.

Zheng, M., X. Ming, L. Wang, D. Yin, and X. Zhang. 2017. "Status Review and Future Perspectives on the Framework of Smart Product Service Ecosystem." *Procedia CIRP* 64: 181–186. https://doi.org/10.1016/j.procir.2017.03.037.

Zhong, X., and G. Ren. 2025. "Digitalization and Firms' Innovation Efficiency: Do Corporate Social Responsibility and Irresponsibility Matter?" *Journal of Technology Transfer* 50: 856–888. https://doi.org/10.1007/s10961-024-10075-9.

Zhou, J., and W. Liu. 2024. "Carbon Reduction Effects of Digital Technology Transformation: Evidence from the Listed Manufacturing Firms in China." *Technological Forecasting and Social Change* 198: 122999. https://doi.org/10.1016/j.techfore.2023.122999.

Zhu, M., C. Liang, P. K. C. Lee, A. C. L. Yeung, and H. Zhou. 2026. "Does Digital Technologies Deployment Promote Environmental Performance? Evidence From China." *IEEE Transactions on Engineering Management* 73: 712–725. https://doi.org/10.1109/TEM.2025.3647213.

Zhu, Q., D. Ma, and X. He. 2023. "Digital Transformation and Firms' Pollution Emissions." *Technological Forecasting and Social Change* 197: 122910. https://doi.org/10.1016/j.techfore.2023.122910.

10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix A

TABLE A1 | Independent and Dependent Variables.

Independent variables (digitalization)	
Artificial intelligence	Software and hardware technology that employs technologies to perform complex tasks and emulate human capabilities through programming and algorithmic processing
Smart product platform	System that integrates advanced technologies and intelligent functions through physical and digital components to offer more connected, adaptable, and sustainable products throughout their life cycle
Digital innovation	Business process digitalization; CRM system; digital technology; digital transformation; digital servitization; digitization; e-commerce; informatization technology; Internet business model; technology and digital economy; web-based accounting
Industrial robotics	Industrial robots and robot applications
Industry 4.0 technologies	Big data; blockchain technology; cloud-based computing; computer-aided design; intelligent manufacturing; internet; machine learning; others
Other digital orientations	Digital orientation (new ways to enhance the effectiveness of digital technology use, experimentation with new digital technologies, adoption of the latest digital technological innovations); digital investment (It reflects a company's expenses on the ongoing acquisition of digital resources such as equipment, technologies, and devices through which companies advance in the digitization process until they achieve digital transformation); digital government (It refers to using digital technology means such as big data and algorithms to drive innovation within governmental departments' internal organizational structures); technological innovation efficiency (a company's capacity for technological innovation to transform relevant inputs into expected results)
Dependent variables (eco-innovation)	
Green innovation	Green technology innovation; green process innovation; green product innovation; general innovation and sustainable innovation; green innovation quality; green innovation quantity; circular eco-innovation; green technology innovation performance; environmental investment in production and operation activities to strengthen the control of environmental pollution; sustainable consumption and production as sustainable management and efficient use of natural resources and reduction of harmful elements. Green innovation as eco-innovation outcomes: corporate environmental performance (company's ability to generate economic value while minimizing environmental impact, consistent with the objectives of eco-innovation); environmental performance (emissions or amount of pollutant discharge); firm performance (within a sustainable development perspective conceptually related to the results of eco-innovation); performance of business technological innovation (measured as expenditures incurred on new products and technologies); innovation efficiency (ability to transform investments in innovation into innovative results); innovation performance (quality improvement through innovation, development of new technologies to improve operational processes, and acquisition of instruments or equipment to accelerate productivity); innovation quality (true value of the innovation process); open innovation (searching for technology, information, and ideas to develop innovative products and services); production efficiency (a company's ability to maximize output without increasing resource inputs)
Decarbonization	Low carbon innovation (cleaner use of traditional energy, improved energy efficiency, the use of new energy sources, and carbon capture, utilization, and storage technologies); Pollution emissions of firm (chemical oxygen demand, sulfur dioxide, ammonia nitrogen, nitrogen oxide, solid waste, industrial wastewater, industrial waste gas, smoke, dust and particulate matter emissions to comprehensively measure the intensity of firms' pollution emissions); carbon emission intensity (direct carbon emissions [coal and oil] and indirect carbon emissions [electricity]); enterprise's carbon emission (ratio of the enterprise's carbon dioxide emissions to the enterprise's primary business income); carbon performance (reduction in energy consumption, carbon emissions and use of materials); low-carbon technology (patents for alternative energy production and energy saving, pollution control in waste management and carbon emissions)

Note: This table shows the independent and dependent variables of the studies selected for the meta-regression analysis as well as their general conceptual definition. The dependent variables mainly capture company outcomes related to eco-innovation, green transition and decarbonization, while the independent variables represent different types of digitalization. Table A5 indicates in detail the variable of each study.

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TABLE A2 | Keywords and Search Strings.

Category	Keywords and synonyms	Search strings
Search String 1: December 2025 Category: Digital technologies Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 3,860,130 results	Digital technologies, Digital transformation, Digital orientation, Industry 4.0, Enabling technology, Fourth Industrial Revolution, Additive manufacturing, 3D printing, Lean 4.0, Smart Manufacturing, Digital manufacturing, Cloud Computing, Machine learning, Artificial intelligence, IoT, Internet of Things, Robot, Smart factories	(((AB = ("Digital technolog*" OR "Digital*" OR "Digital transformation" OR "Digital orientation" OR "Industry 4.0" OR "Enabling technolog*" OR "Fourth Industrial Revolution" OR "Additive manufacturing" OR "3D printing" OR "Lean 4.0" OR "Smart Manufacturing" OR "Digital manufacturing" OR "Cloud Computing" OR "Machine learning" OR "Artificial intelligence" OR "IoT" OR "Internet of Things" OR "Robot*" OR "Smart factories") OR TI = ("Digital technolog*" OR "Digital*" OR "Digital transformation" OR "Digital orientation" OR "Industry 4.0" OR "Enabling technolog*" OR "Fourth Industrial Revolution" OR "Additive manufacturing" OR "3D printing" OR "Lean 4.0" OR "Smart Manufacturing" OR "Digital manufacturing" OR "Cloud Computing" OR "Machine learning" OR "Artificial intelligence" OR "IoT" OR "Internet of Things" OR "Robot*" OR "Smart factories") OR AK = ("Digital technolog*" OR "Digital*" OR "Digital transformation" OR "Digital orientation" OR "Industry 4.0" OR "Enabling technolog*" OR "Fourth Industrial Revolution" OR "Additive manufacturing" OR "3D printing" OR "Lean 4.0" OR "Smart Manufacturing" OR "Digital manufacturing" OR "Cloud Computing" OR "Machine learning" OR "Artificial intelligence" OR "IoT" OR "Internet of Things" OR "Robot*" OR "Smart factories")))
Search String 2: December 2025 Category: Eco-innovation Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 3,149,474 results	Eco-innovation, decarbonization, Net-zero transition, Circular economy, Cleaner technology, Environmental, Green innovation, Sustainable development, Energy conservation, Emission reduction and Low-carbon	(((AB = ("Eco-innovat*" OR "Decarbonization" OR "Net-zero transition" OR "Circular economy" OR "Cleaner technolog*" OR "Environmental" OR "Green innov*" OR "Sustainable development" OR "Energy conservation" OR "Emission reduction" OR "Low-carbon") OR TI = ("Eco-innovat*" OR "Decarbonization" OR "Net-zero transition" OR "Circular economy" OR "Cleaner technolog*" OR "Environmental" OR "Green innov*" OR "Sustainable development" OR "Energy conservation" OR "Emission reduction" OR "Low-carbon") OR AK = ("Eco-innovat*" OR "Decarbonization" OR "Net-zero transition" OR "Circular economy" OR "Cleaner technolog*" OR "Environmental" OR "Green innov*" OR "Sustainable development" OR "Energy conservation" OR "Emission reduction" OR "Low-carbon"))))
Search String 3: December 2025 Category: Twin transition Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 19,943 results	Green digital transition, Green transition, New green IT, Green technology, Green-tech-enabling, Environmental technology and Twin transition	((AB = ("Green digital transition" OR "Green transition" OR "New green IT" OR "Green technolog*" OR "Green-tech-enabling" OR "Environmental technolog*" OR "Twin transition") OR TI = ("Green digital transition" OR "Green transition" OR "New green IT" OR "Green technolog*" OR "Green-tech-enabling" OR "Environmental technolog*" OR "Twin transition") OR AK = ("Green digital transition" OR "Green transition" OR "New green IT" OR "Green technolog*" OR "Green-tech-enabling" OR "Environmental technolog*" OR "Twin transition")))
Search String 4: December 2025 Category: Combining categories 1 (Digital technologies) and 2 (Eco-innovation) using the Boolean operator "and" Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 141,785 results		
Search String 5: December 2025 Category: Combining categories 3 (Twin transition) and 4 using the Boolean operator "or" Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 160,213 results		

(Continues)

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TABLE A2 | (Continued)

Category	Keywords and synonyms	Search strings
Search String 6: December 2025 Category: Journals of the chosen categories Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 330,403 results	Academic Journal Guide (AJG) 2021 ranking, selecting only journals with rankings 3, 4, and 4*, except in the field of OPS&TECH in which, given its nature, journals with rankings from 2 to 4* were included. The fields to which the journals belonged were: Administration, Corporate Social Responsibility and Ethics (ETHICS-CSR-MAN), Administration and Information (INFO MAN), Innovation (INNOV), Operations and Technology (OPS&TECH), Organization and Management Science (OR&MANSCI), Regional Studies (REGIONAL STUDIES), Planning and Environment (PLANNING AND ENVIRONMENT), and Strategy (STRAT)	SO = ("Academy of Management Annals" OR "Academy of Management Journal" OR "Academy of Management Review" OR "Administrative Science Quarterly" OR "Journal of Management" OR "Academy of Management Perspectives" OR "British Journal of Management" OR "Business Ethics Quarterly" OR "Journal of Management Studies" OR "Academy of Management Discoveries" OR "Business and Society" OR "California Management Review" OR "European Management Review" OR "Gender and Society" OR "Gender, Work and Organization" OR "Harvard Business Review" OR "International Journal of Management Reviews" OR "Journal of Business Ethics" OR "Journal of Business Research" OR "Journal of Management Inquiry" OR "MIT Sloan Management Review" OR "Information Systems Research" OR "Journal of the Association for Information Systems" OR "MIS Quarterly: Management Information Systems" OR "European Journal of Information Systems" OR "Information Systems Journal" OR "Journal of Information Technology" OR "Journal of Management Information Systems" OR "Journal of Strategic Information System" OR "Decision Support Systems" OR "Government Information Quarterly" OR "Information and Management" OR "Information and Organization" OR "Information Society" OR "Information Systems Frontiers" OR "Information Technology and People" OR "International Journal of Electronic Commerce" OR "Internet Research" OR "Journal of Computer-Mediated Communication" OR "Journal of the Association for Information Science and Technology" OR "Research Policy" OR "Journal of Product Innovation Management" OR "Industry and Innovation" OR "Journal of Technology Transfer" OR "R and D Management" OR "Technological Forecasting and Social Change" OR "Technovation" OR "Journal of Operations Management" OR "International Journal of Operations and Production Management" OR "Journal of Supply Chain Management" OR "Production and Operations Management" OR "Computers in Industry" OR "IEEE Transactions on Engineering Management" OR "International Journal of Production Economics" OR "International Journal of Production Research" OR "Journal of Business Logistics" OR "Journal of Purchasing and Supply Management" OR "Manufacturing and Service Operations Management" OR "Production Planning and Control" OR "Supply Chain Management" OR "Business Process Management Journal" OR "Computers and Industrial Engineering" OR "International Journal of Computer Integrated Manufacturing" OR "International Journal of Physical Distribution and Logistics Management" OR "International Journal of Project Management" OR "International Journal of Quality and Reliability Management" OR "International Journal of Technology Management" OR "Journal of Construction Engineering and Management—ASCE" OR "Journal of Scheduling" OR "Total Quality Management and Business Excellence" OR "Management Science" OR "Operations Research" OR "European Journal of Operational Research" OR "IEEE Transactions on Evolutionary Computation" OR "Mathematical Programming" OR "ACM Transactions on Modeling and Computer Simulation" OR "Annals of Operations Research" OR "Computational Optimization and Applications" OR "Computers and Operations Research" OR "Decision Sciences" OR "Evolutionary Computation" OR "Fuzzy Optimization and Decision Making" OR "IEEE Transactions on Cybernetics" OR "IEEE Transactions on Systems, Man, and Cybernetics: Systems" OR "IIE Transactions" OR "INFORMS Journal on Computing" OR "International Journal of Forecasting" OR "Journal of Heuristics" OR "Journal of Optimization Theory and Applications" OR "Journal of the Operational Research Society" OR "Mathematics of Operations Research" OR "Naval Research Logistics" OR "Omega" OR "OR Spectrum" OR "Reliability Engineering and System Safety" OR "SIAM Journal on Optimization" OR "Transportation Science" OR "Economic Geography" OR "Journal of Economic Geography" OR "Regional Studies" OR "Cambridge Journal of Regions, Economy and Society" OR "Environment and Planning A" OR "European Urban and Regional Studies" OR "Global Environmental Change" OR "Journal of Regional Science" OR "Journal of Rural Studies" OR "Papers in Regional Science" OR "Regional Science and Urban Economics" OR "Urban Studies" OR "Strategic Management Journal" OR "Global Strategy Journal" OR "Long Range Planning" OR "Strategic Organization")

(Continues)

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TABLE A2 | (Continued)

Category	Keywords and synonyms	Search strings
Search String 7: December 2025 Category: Combining categories 5 and 6 using the Boolean operator "and" Type of Documents: Only published articles (both academics and practitioners) Language: English Time period: All years Web of Science: 1450 results		

Note: All searches were conducted in the Web of Science Core Collection (until the end of December 2025), restricted to peer-reviewed, practitioner-oriented articles published in English, with no time limit. Search categories included (1) digital technologies, (2) eco-innovation, and (3) twin transition, as well as their Boolean combinations. A journal filter, applied using the Scholarly Journal Guide (AJG 2021), restricted the sample to journals with a ranking of 3, 4, or 4* (and 2 to 4* in Operations and Technology). The search strategy yielded 1450 potentially relevant articles.

TABLE A3 | Articles by journal and country of publication.

Journal	Number of articles	Country where study was conducted	Number of articles
Technological Forecasting and Social Change	10	China	26
International Journal of Production Economics	6	USA	3
IEEE Transactions on Engineering Management	6	India	2
Journal of Technology Transfer	3	Europe	1
Annals of Operations Research	2	Ghana	1
Journal of Business Research	2	Italy	1
Technovation	2	Korea	1
British Journal of Management	1	Pakistan	1
Business Process Management Journal	1	Spain	1
Industry and Innovation	1	United Kingdom	1
Information and Management	1	**TOTAL**	**38**
Information Systems Frontiers	1		
Internet Research	1		
Regional Studies	1		
TOTAL	**38**		

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TABLE A4 | Examples of Articles Excluded From the Systematic Review.

Citation	Title	DOI	Reason for exclusion
Gao et al. (2024)	Deterrent Effects of Central Environmental Protection Inspection on Green Technology Innovation Cycles: Evidence From the Listed State-Owned Firms in China	https://10.1016/j.techfore.2024.123783	The independent variable is not consistent with digitalization.
Haleem et al. (2024)	Driving Organizational Value Through Sustainable Business Processes and Green Information Technology	10.1108/BPMJ-01-2024-0011	The dependent variable is not consistent with eco-innovation.
Hameed et al. (2023)	Ethical Leadership and Environmental Performance: The Role of Green IT Capital, Green Technology Innovation, and Technological Orientation	10.1016/j.techfore.2023.122739	The independent variable is not consistent with digitalization.
Li et al. (2024)	Unveiling the Green Innovation Paradox: Exploring the Impact of Carbon Emission Reduction on Corporate Green Technology Innovation	10.1016/j.techfore.2024.123562	The independent variable is not consistent with digitalization.
Lin and Xie (2024)	Driving Green Technology Innovation in Renewable Energy: Does Venture Capital Matter?	10.1109/TEM.2023.3286978	The independent variable is not consistent with digitalization.
Lubacha and Wendler (2021)	Do European Firms Obey the Rules? Environmental Innovativeness in Light of Institutional Frameworks	10.1080/13662716.2021.1929869	The dependent variable is not consistent with eco-innovation.
Ning and Guo (2022)	Technological Diversification to Green Domains: Technological Relatedness, Invention Impact and Knowledge Integration Capabilities	10.1016/j.respol.2021.104406	The independent variable is not consistent with digitalization, nor is the dependent variable consistent with eco-innovation.
Opazo-Basáez et al. (2024)	Evaluating the Effect of Green Technological Innovations on Organizational and Environmental Performance: A Treble Innovation Approach	10.1016/j.technovation.2023.102885	The independent variable is not consistent with digitalization, nor is the dependent variable consistent with eco-innovation.
Sheng et al. (2024)	Corporate Green Innovation in an Aging Population: Evidence From Chinese Listed Companies.	10.1016/j.techfore.2024.123307	The independent variable is not consistent with digitalization.
Turkcan et al. (2022)	To Be More Innovative and More Competitive in Dynamic Environments: The Role of Additive Manufacturing	10.1016/j.ijpe.2022.108418	The dependent variable is not consistent with eco-innovation.

Note: A complete list of all articles identified by the search process is available on request.

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TABLE A5 | Bibliographic Sources, Variables, and Conceptual Definitions Used in the Meta-Regression Analysis.

Bibliography	Variables and conceptual definitions
Ardito et al. (2021). The Duality of Digital and Environmental Orientations in the Context of SMEs: Implications for Innovation Performance. Journal of Business Research, 123, 44–56. https://doi.org/10.1016/j.jbusres.2020.09.022	Digital orientation, in the context of other digital orientations, is measured by counting the number of functional areas of the organization in which digital technologies have been adopted and their effect on product and process innovation
Bendig et al. (2023). Digital Orientation and Environmental Performance in Times of Technological Change. Technological Forecasting and Social Change, 188. https://doi.org/10.1016/j.techfore.2022.122272	Digital orientation, in the context of other digital orientations, covers advanced communications and technologies, organizational digital capabilities, coordination of the digital ecosystem and the configuration of the digital architecture in the company, and its effect on environmental performance, understood as the reduction of emissions, waste, effluents and waste, especially with regard to CO_2 emissions and total waste, is also measured by product innovation, especially in terms of minimizing the cost of its life cycle as it is related to sustainable development and, finally, environmental performance is also measured by the appropriate use of resources, particularly water and energy
Cattani et al. (2025). Firms' Eco-Innovation and Industry 4.0 Technologies in Urban and Rural Areas. Regional Studies, 59(1). https://doi.org/10.1080/00343404.2023.2243984	Industry 4.0 technologies are those listed by the Eurobarometer (artificial intelligence, cloud computing, robotics, smart devices, big data analytics, high-speed infrastructure and blockchain) that affect eco-innovation, which can be understood as the production, assimilation or exploitation of a product, production process, service or management or business methods that is novel to the company [or organization] and that results, throughout its life cycle, in a reduction of environmental risk, pollution and other negative impacts of resource use (including energy use) compared to relevant alternatives
Chin et al. (2021). The Moderating Effect of Managerial Discretion on Blockchain Technology and the Firms' Innovation Quality: Evidence From Chinese Manufacturing Firms. International Journal of Production Economics, 240. https://doi.org/10.1016/j.ijpe.2021.108219	Blockchain technology, in the context of Industry 4.0 technologies, and its effect on the quality of the innovation, which is understood as adequate when the value of the product and the value of the innovation activities is greater after the innovation process
Dong et al. (2024). Can Corporate Digital Innovation Speed Up Climate Change Mitigation? Evidence From China. IEEE Transactions on Engineering Management, 71, 7797–7814. https://doi.org/10.1109/TEM.2024.3386181	Technology and the digital economy, in the context of digital innovation, refer to the integration of information, computing, communication and connectivity technologies by companies within the innovation process, which encompasses the introduction of products, improvements in production processes, changes in organizational models and the creation and alteration of business models and their effect on the intensity of carbon emissions measured as the ratio between aggregate carbon emissions (measured in metric tons) and operating revenues
Fosso Wamba, Queiroz, Pappas, and Sullivan (2024). Artificial Intelligence Capability and Firm Performance: A Sustainable Development Perspective by the Mediating Role of Data-Driven Culture. Information Systems Frontiers, 26, 2189–2203. https://doi.org/10.1007/s10796-023-10460-z	Artificial intelligence and its effect on the firm performance, from a sustainable development perspective that links organizational results with environmentally oriented innovation practices
Hagan et al. (2024). A Moderated Model of Open Innovation: An Integration of Theories of Constraint and Dynamic Capabilities. IEEE Transactions on Engineering Management, 71, 14,438–14,449. https://doi.org/10.1109/TEM.2024.3440070	Digital transformation, in the context of digital innovation, is understood as the activities that drive the company towards new processes based on technologies, and its effect on open innovation understood as the search for technology, information and ideas that complement internal knowledge so that value can be created for the company and innovative products and services can be developed
Hao et al. (2024). The Impact of Digital Government on Corporate Green Innovation: Evidence From China. Technological Forecasting and Social Change, 206. https://doi.org/10.1016/j.techfore.2024.123570	Digital government, in the context of other digital orientations, refers to using digital technology means such as big data and algorithms to drive innovation within governmental departments' internal organizational structures and its impact on green innovation

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10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE A5 | (Continued)

Bibliography	Variables and conceptual definitions
Huang et al. (2026). Cutting Emissions Through Intelligent Production in Chinese Manufacturing Firms: An Empirical Analysis. Annals of Operations Research, 359, 1849–1872. https://doi.org/10.1007/s10479-024-06005-5	Robot applications, in the context of industrial robotics, refers to the operational functions of robots and their effect on the intensity of carbon emissions measured as direct carbon emissions (coal and oil) and indirect carbon emissions (electricity) as well as on the performance of business technological innovation, the latter measured as expenses incurred on new products and technologies
Jibril et al. (2024). Do Digital Technologies Enable Firms that Prioritize Sustainability Goals to Innovate? Empirical Evidence From Established UK Micro-Businesses. British Journal of Management, 35(4), 2155–2173. https://doi.org/10.1111/1467-8551.12821	CRM systems, e-commerce, web-based accounting in the context of digital innovation, computer-aided design, cloud computing, machine learning in the context of Industry 4.0 technologies and artificial intelligence and their effect on product and process innovation.
Kong and Feng (2025). Linking Digital Transformation With Supply Chain Intelligence Integration and Carbon Performance: Does Business Model Innovation Matter? Business Process Management Journal, 31(4), 1274–1305. https://doi.org/10.1108/BPMJ-09-2023-0682	Digital transformation, in the context of digital innovation, can be defined as a process aimed at improving organizational effectiveness through the integration of digital technologies to drive significant changes in business strategy, activities, processes, and their effect on carbon performance understood as the reduction in energy consumption, carbon emissions and the use of materials
Lee et al. (2024). Exploring the Role of Digital Servitization for Green Innovation: Absorptive Capacity, Transformative Capacity, and Environmental Strategy. Technological Forecasting and Social Change, 207. https://doi.org/10.1016/j.techfore.2024.123614	Digital servitization (divided into two dimensions: digitalization and servitization), in the context of digital innovation, and its impact on green innovation
Lin et al. (2024). How Does Artificial Intelligence Affect the Environmental Performance of Organizations? The Role of Green Innovation and Green Culture. Information and Management, 61(2). https://doi.org/10.1016/j.im.2024.103924	Artificial intelligence that impacts green product innovation and green process innovation
Liu et al. (2025). Exploring Drivers of Eco-Innovation in Manufacturing Firms' Circular Economy Transition: An Awareness, Motivation, Capability Perspective. Annals of Operations Research, 355, 147–182. https://doi.org/10.1007/s10479-023-05473-5	Technological innovation efficiency, in the context of other digital orientations, is the technological innovation capacity of a company to transform relevant inputs into expected results and its effect on circular eco-innovation
Martínez Falcó et al. (2025). Digital Transformation and Green Innovation Performance: Unraveling the Role of Green Knowledge Sharing and Top Management Environmental Awareness. Internet Research, 35(4), 1621–1656. https://doi.org/10.1108/INTR-11-2023-1016	Digital transformation, in the context of digital innovation, refers to the company's practices to digitize as much as possible, collect large amounts of data from various sources, create greater interconnection between different business processes through digital technologies, and improve the efficiency of the customer interface, and its effect on green innovation performance
Mondal et al. (2024). Achieving Technological Transformation and Social Sustainability: An Industry 4.0 Perspective. IEEE Transactions on Engineering Management, 71, 6623–6635. https://doi.org/10.1109/TEM.2023.3246638	Digital transformation, in the context of digital innovation, refers to the alteration or adaptation of business models as a consequence of dynamic technological progress and innovations that drive changes in sustainable businesses and society as a whole, and their effect on sustainable consumption and production, understood as the sustainable management and efficient use of natural resources and the reduction of harmful elements
Montresor and Vezzani (2023). Digital Technologies and Eco-Innovation. Evidence of the Twin Transition From Italian Firms. Industry and Innovation, 30(7), 766–800. https://doi.org/10.1080/13662716.2023.2213179	Digital technologies are the areas of application of AI (elaboration and analysis of big data, interactive, immersive technologies + advanced automation, collaborative robots and smart systems), the internet based technologies (Internet of things, internet connection) and other technological areas (3D printing, security and simulation [simulation of interconnected machines + cyber-security]) that have an effect on eco-innovation

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TABLE A5 | (Continued)

Bibliography	Variables and conceptual definitions
Ranjan (2024). Unraveling the Mystery of the Link Between Digital Orientation and Innovation Performance: The Interplay of Digital Business Capability and Environmental Dynamism. Technovation, 131. https://doi.org/10.1016/j.technovation.2024.102966	Digital orientation, in the context of other digital orientations, is a business-level concept that represents a company's strategic decision to adopt emerging digital technologies in response to changes in the technological environment and their effect on innovation performance, understood as the ability to develop new technologies to improve operational processes, as well as the acquisition of technology to accelerate productivity
Shahzad et al. (2026). Integrating Industry 4.0 Technologies Into Green Supply Chains: An Empirical Investigation of Pathways to Green Innovation and Circular Economy. International Journal of Production Economics, 291. https://doi.org/10.1016/j.ijpe.2025.109859	Industry 4.0 technologies refer to the use of digital technologies to achieve the effective integration of environmentally sustainable practices in the supply chain and their impact on green process innovation and green product innovation
Shao et al. (2024). Leveraging Corporate Digitalization for Green Technology Innovation: The Mediating Role of Resource Endowments. Technovation, 133. https://doi.org/10.1016/j.technovation.2024.102999	Digitization, in the context of digital innovation, includes technologies such as augmented reality, virtual reality, the industrial Internet, smart manufacturing, voice recognition, biometrics, identity authentication, natural language processing, among others, and its effect on the innovation of green technologies.
Song et al. (2024). Digital Transformation and Corporate Environmental Performance: Evidence From Chinese Listed Companies. Technological Forecasting and Social Change, 201. https://doi.org/10.1016/j.techfore.2023.123159	Digital transformation, informatization technology, internet business model (in the context of digital innovation) or big data and intelligent manufacturing (in the context of Industry 4.0 technologies) and its effect on the corporate environmental performance as an indirect measure of emissions
Tao et al. (2025). Sensing and Seizing Opportunities for Green Innovation: The Role of Business Process Digitalization. IEEE Transactions on Engineering Management, 72, 2798–2810. https://doi.org/10.1109/TEM.2025.3583782	Business process digitalization, in the context of digital innovation, refers to the use of digital technologies to support business activities related to supplier relationships, product and service development activities, production and operations, and their effect on green product innovation and green process innovation
Tian et al. (2023). Can Enterprise Green Technology Innovation Performance Achieve "Corner Overtaking" by Using Artificial Intelligence?—Evidence From Chinese Manufacturing Enterprises. Technological Forecasting and Social Change, 194. https://doi.org/10.1016/j.techfore.2023.122732	AI uses machine learning, deep learning, computer vision, and other technologies to emulate humans through programming and algorithm processing, impacting innovation performance in enterprise green technology
Wang, Gong, and Yang (2024). Unveiling the Relation Between Digital Technology and Low-Carbon Innovation: Carbon Emission Trading Policy as an Antecedent. Technological Forecasting and Social Change, 205. https://doi.org/10.1016/j.techfore.2024.123522	Digital technology, in the context of digital innovation signifies companies' digital transformation and applications of digital technology, and its effect on the low carbon innovation understood as the cleaner use of traditional energy, improved energy efficiency, the use of new energy sources, and carbon capture, utilization, and storage technologies.
Wang et al. (2025). The Impact of Artificial Intelligence on Corporate Environmental Investment. IEEE Transactions on Engineering Management, 72, 96–114. https://doi.org/10.1109/TEM.2024.3491940	Artificial intelligence and its effect on environmental investment, which refers to expenditures directly related to environmental protection, such as desulfurization and denitrification projects, wastewater treatment and initiatives for waste gas management, dust removal and energy saving
Wang et al. (2023). Robots and Firm Innovation: Evidence From Chinese Manufacturing. Journal of Business Research, 162. https://doi.org/10.1016/j.jbusres.2023.113879	Industrial robots, in the context of industrial robotics, are defined as multifunctional, reprogrammable, and automatically controlled machines that promote clean production and adopt eco-efficient ways of using resources and materials, and their effect on general innovation and sustainable innovation
Xu et al. (2023). Making Technological Innovation Greener: Does Firm Digital Transformation Work? Technological Forecasting and Social Change, 197. https://doi.org/10.1016/j.techfore.2023.122928	Digital transformation, in the context of digital innovation, includes artificial intelligence technology, blockchain technology, cloud computing technology, big data technology, and digital technology applications such as B2B, B2C, C2B, and C2C and its effect on green technological innovation

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10990836, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bse.71801 by Enrique Acebo - Bucle - Universidad De Leon , Wiley Online Library on [10/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE A5 | (Continued)

Bibliography	Variables and conceptual definitions
Yang et al. (2023). Digital Transformation and Low-Carbon Technology Innovation in Manufacturing Firms: The Mediating Role of Dynamic Capabilities. International Journal of Production Economics, 263. https://doi.org/10.1016/j.ijpe.2023.108969	Digital transformation, in the context of digital innovation, is the adoption of technological practices and their effect on low carbon technology understood as the production of alternative energies and energy savings, pollution control, waste management and the reduction of carbon emissions.
Yang and Luo (2025). Bridging Digital Transformation and Green Innovation in Firms: A Dual-Dimensional Analysis of Disclosure-Behavior Alignment. The Journal of Technology Transfer. https://doi.org/10.1007/s10961-025-10278-8	Digital transformation, in the context of digital innovation, refers both to the underlying technologies and the practical applications of digital technology, and its effect on the quantity and quality of green innovation
Ye et al. (2023). Digital Investment and Environmental Performance: The Mediating Roles of Production Efficiency and Green Innovation. International Journal of Production Economics, 259. https://doi.org/10.1016/j.ijpe.2023.108822	Digital investment, in the context of other digital orientations, refers to the advancement of companies towards the digitalization process through the continuous purchase of digital technologies and devices to finally achieve digital transformation, and its effect on green innovation, as well as on environmental performance (emissions) and productive efficiency
Zhang et al. (2026). Digital Government Development, Corporate Digital Transformation, and Carbon Performance: Insights From Listed Companies in China. Technological Forecasting and Social Change, 223. https://doi.org/10.1016/j.techfore.2025.124414	Digital government, in the context of other digital orientations, is understood as digital service capacity in terms of service delivery, responsiveness, and service intelligence capabilities, and its effect on carbon performance measured in terms of carbon emissions
Zhang and Deborah (2025). Digitalization, Internationalization and Green Innovation in China: An Analysis Based on Threshold and Mediation Effects. Journal of Technology Transfer, 50, 962–994. https://doi.org/10.1007/s10961-024-10116-3	Digitization, in the context of digital innovation, (blockchain technology and digital currency; business intelligence and machine learning; data mining and text mining; such as the Internet of things and cloud computing and the application of technologies such as B2B and B2C) impacts on enterprise green innovation
Zhang et al. (2024). Smart Platforming in Automotive Manufacturing for NetZero: Intelligentization, Green Technology, and Innovation Dynamics. International Journal of Production Economics, 274. https://doi.org/10.1016/j.ijpe.2024.109289	Smart product platform that integrates customers, smart product service systems, smart service platform and product service providers for value co-creation and customer experience enhancement, through intelligent interaction, mutual cooperation, resource sharing and optimal configuration that impacts green technology innovation.
Zheng et al. (2024). Is It Possible for Semiconductor Companies to Reduce Carbon Emissions Through Digital Transformation? Evidence From China. International Journal of Production Economics, 272. https://doi.org/10.1016/j.ijpe.2024.109246	Digital transformation, in the context of digital innovation, is the use of digitization by the company and its effect on carbon emissions intensity, which are the estimated emissions (based on the industry's energy consumption) of carbon dioxide in relation to operating income
Zhong and Ren (2025). Digitalization and Firms' Innovation Efficiency: Do Corporate Social Responsibility and Irresponsibility Matter? Journal of Technology Transfer, 50, 856–888. https://doi.org/10.1007/s10961-024-10075-9	Digitalization, in the context of digital innovation, refers to artificial intelligence, blockchain, cloud computing, and big data technologies, and their effect on innovation efficiency, understood as the ability to transform investments in innovation into innovative results
Zhou and Liu (2024). Carbon Reduction Effects of Digital Technology Transformation: Evidence From the Listed Manufacturing Firms in China. Technological Forecasting and Social Change, 198. https://doi.org/10.1016/j.techfore.2023.122999	Digital transformation, in the context of digital innovation, understood as the introduction of advanced technologies in the production and management of companies to optimize production and business services, and its effect on enterprise's carbon dioxide emission
Zhu et al. (2026). Does Digital Technologies Deployment Promote Environmental Performance? Evidence From China. IEEE Transactions on Engineering Management, 73, 712–725. https://doi.org/10.1109/TEM.2025.3647213	Digital technology, in the context of digital innovation, refers to the underlying technologies, as well as the application of digital technologies and their effect on environmental performance, which refers to environmental strengths that include environmental protection practices, the establishment of mechanisms to respond to environmental emergencies, and environmental concerns that encompass actions to control adverse environmental events

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TABLE A5 | (Continued)

Bibliography	Variables and conceptual definitions
Zhu et al. (2023). Digital Transformation and Firms' Pollution Emissions. Technological Forecasting and Social Change, 197. https://doi.org/10.1016/j.techfore.2023.122910	Digital transformation in the context of digital innovation includes smart manufacturing, smart logistics, artificial intelligence, data mining, robotics, the Internet, big data, e-commerce, cloud computing, among others, and its effect on companies' polluting emissions, understood as chemical oxygen demand, sulfur dioxide, ammonia nitrogen, nitrogen oxide, solid waste, industrial wastewater, industrial waste gases, smoke, dust, and particulate emissions to comprehensively measure the intensity of companies' polluting emissions

Note: This table presents the empirical studies included in the meta-regression analysis and summarizes the conceptual definitions of the variables employed in the reviewed literature, thereby ensuring comparability and consistency across studies.

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TABLE A6 | Summary Statistics.

		Unweighted			Weighted		
	Observations	**Mean**	**95% conf. int.**		**Mean**	**95% conf. int.**	
Type of digitalization							
Artificial intelligence	77	0.091	0.062	0.120	0.024	0.018	0.030
Digital innovation	338	0.021	0.011	0.031	0.026	0.021	0.030
Industrial robotics	38	0.002	−0.003	0.007	0.006	0.003	0.010
Industry 4.0 technologies	35	0.059	−0.001	0.118	0.010	0.004	0.016
Other digital orientations	71	0.095	0.075	0.115	0.037	0.025	0.049
Smart product platform	28	0.141	0.092	0.190	0.14	0.093	0.187
Regional effects							
China	525	0.037	0.028	0.045	0.008	0.006	0.009
Ghana	4	0.487	0.449	0.526	0.488	0.449	0.527
India	5	0.264	0.243	0.284	0.262	0.240	0.284
Italy	15	0.017	0.007	0.027	0.016	0.008	0.025
Pakistan	6	0.223	0.123	0.322	0.226	0.127	0.325
United Kingdom	14	0.022	0.007	0.036	0.022	0.007	0.036
USA	12	0.152	0.101	0.202	0.108	0.065	0.152
Various	6	0.111	−0.031	0.252	0.044	0.018	0.069
Estimation characteristics							
Control gas emissions	15	0.017	0.007	0.027	0.016	0.008	0.025
Control R&D	165	0.055	0.042	0.067	0.026	0.023	0.030
Environmental regulation	38	0.075	0.059	0.090	0.029	0.017	0.040
External collaboration	33	0.042	0.018	0.065	0.017	0.011	0.023
Control firm exporting	66	0.012	0.007	0.017	0.007	0.004	0.010
Control ownership type	348	0.046	0.038	0.055	0.016	0.014	0.017
Control external financing	29	0.019	0.011	0.027	0.017	0.011	0.022
Control firm growth	101	0.025	0.008	0.042	0.023	0.016	0.031
Control age	328	0.034	0.022	0.047	0.015	0.013	0.016
Control size	409	0.037	0.027	0.047	0.018	0.015	0.021
Model type							
Probit or logit	49	0.072	0.046	0.098	0.019	0.012	0.025
Structural equation modeling	23	0.328	0.256	0.401	0.344	0.265	0.423
Ordinary least squares	152	0.068	0.051	0.085	0.015	0.013	0.017
Instrumental variable	73	0.027	0.017	0.038	0.003	−0.000	0.007
Panel data with fixed effects	290	0.012	0.003	0.21	−0.001	−0.003	0.000
Data characteristics							
Longitudinal data	535	0.031	0.024	0.038	0.008	0.007	0.009
Cross sectional data	52	0.198	0.419	0.248	0.054	0.028	0.079
All estimates	***587***	***0.046***	***0.037***	***0.055***	***0.008***	***0.007***	***0.009***

Note: This table includes the subsample-specific means for the estimated effects of digitalization on eco-innovation, weighted by the inverse of the variance of effect size. The "various" category includes countries in Europe as well as Korea and Spain specifically.

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TABLE A7 | Publication Bias Tests for Technology Dimension Subsets.

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Block 1: Artificial intelligence							
Publication bias	3.438***	−1.958**	3.218***	1.930***	3.169***		1.930***
(Standard error)	(0.519)	(0.979)	(0.653)	(0.561)	(0.526)		(0.504)
	[2.422 to 4.455]	[−3.910 to −0.006]	[1.939 to 4.497]	[0.831 to 3.030]	[2.137 to 4.200]		
Effect beyond bias	−0.007	0.147***	−0.001	0.014**	0.001	0.023***	0.014***
(Standard error)	(0.019)	(0.029)	(0.022)	(0.006)	(0.005)	(0.002)	(0.004)
	[−0.044 to 0.300]	[0.090 to 0.204]	[−0.043 to 0.042]	[0.003 to 0.025]	[−0.010 to 0.011]		
Observations	77	77	77	77	77	35	77
Studies	6	6	6	6	6	2	6
Block 2: Digital innovation							
Publication bias	0.131	0.423*	−0.529	−0.428	−0.554		−0.428
(Standard error)	(0.313)	(0.233)	(0.424)	(0.474)	(0.539)		(0.487)
	[−0.482 to 0.744]	[−0.036 to 0.882]	[−1.360 to 0.302]	[−1.357 to 0.502]	[−1.610 to 0.503]		
Effect beyond bias	0.019**	0.014***	0.030***	0.029***	0.031***	0.023***	0.029***
(Standard error)	(0.007)	(0.005)	(0.009)	(0.004)	(0.005)	(0.002)	(0.005)
	[0.004 to 0.033]	[0.005 to 0.023]	[0.013 to 0.048]	[0.021 to 0.038]	[0.020 to 0.041]		
Observations	338	338	338	338	338	149	338
Studies	20	20	20	20	20	8	20
Block 3: Industrial robotics							
Publication bias	−6.649*	−1.310	−16.277***	−28.633***	−16.266		−28.632***
(Standard error)	(3.752)	(3.414)	(5.159)	(9.714)	(9.958)		(6.898)
	[−14.00 to 0.706]	[−8.241 to 5.621]	[−26.387 to −6.166]	[−47.67 to −9.59]	[−35.78 to −3.25]		
Effect beyond bias	0.008*	0.003	0.018***	0.028***	0.018**	0.006***	0.028***
(Standard error)	(0.043)	(0.004)	(0.006)	(0.007)	(0.008)	(0.002)	(0.005)
	[−0.000 to 0.017]	[−0.005 to 0.011]	[0.007 to 0.029]	[0.014 to 0.041]	[0.003 to 0.032]		
Observations	38	38	38	38	38	37	38
Studies	2	2	2	2	2	2	2
Block 4: Industry 4.0 technologies							
Publication bias	0.659	−15.942***	1.592	2.153***	1.576		2.153***
(Standard error)	(0.804)	(0.577)	(1.214)	(0.769)	(1.037)		(0.757)
	[−0.916 to 2.234]	[−27.78 to −4.108]	[−0.788 to 3.972]	[0.645 to 3.661]	[−0.457 to 3.608]		
Effect beyond bias	0.037	0.581***	0.006	0.003	0.007	0.008*	0.003
(Standard error)	(0.039)	(0.191)	(0.049)	(0.004)	(0.006)	(0.003)	(0.004)
	[−0.039 to 0.113]	[0.190 to 0.972]	[−0.090 to 0.103]	[−0.005 to 0.011]	[−0.004 to 0.018]		

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TABLE A7 | (Continued)

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Observations	35	35	35	35	35	8	35
Studies	6	6	6	6	6	1	6
Block 5: Other digital orientations							
Publication bias	1.827***	−2.104***	2.794***	3.240***	2.745***		3.240***
(Standard error)	(0.531)	(0.664)	(0.638)	(0.468)	(0.419)		(0.498)
	[0.786 to 2.869]	[−3.431 to −0.777]	[1.544 to 4.043]	[2.321 to 4.158]	[1.924 to 3.565]		
Effect beyond bias	0.036*	0.163***	0.005	−0.003	0.007	0.021***	−0.003
(Standard error)	(0.019)	(0.022)	(0.022)	(0.008)	(0.009)	(0.005)	(0.008)
	[−0.002 to 0.074]	[0.119 to 0.206]	[−0.039 to 0.049]	[−0.019 to 0.013]	[−0.010 to 0.024]		
Observations	71	71	71	71	71	16	71
Studies	7	7	7	7	7	2	7
Block 6: Smart product platform							
Publication bias	−0.088	−0.088	0.306	0.061	0.078		0.062
(Standard error)	(0.818)	(0.849)	(0.877)	(1.149)	(0.912)		(1.934)
	[−1.692 to 1.515]	[−1.834 to 1.657]	[−1.412 to 2.024]	[−2.191 to 2.315]	[−1.709 to 1.866]		
Effect beyond bias	0.145***	0.145***	0.126***	0.138**	0.137**	0.138***	0.138*
(Constant)	(0.045)	(0.047)	(0.048)	(0.054)	(0.047)	(0.027)	(0.068)
	[0.056 to 0.234]	[0.048 to 0.242]	[0.033 to 0.220]	[0.031 to 0.244]	[0.046 to 0.229]		
Observations	28	28	28	28	28	20	28
Studies	1	1	1	1	1	1	1

Note: Reported coefficients correspond to the publication bias tests, applied to data subsets classified by digital technology type. Robust standard errors are shown in parentheses, and 95% confidence intervals are reported in brackets. The "publication bias" rows capture the magnitude of the estimated bias associated with the likelihood of publication. The "effect beyond bias" (constant) rows present the average effect corrected for such bias.
***$p < 0.01$.
**$p < 0.05$.
*$p < 0.10$.

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TABLE A8 | Publication Bias Tests for Eco-Innovation Subsets.

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Block 1: Green innovation							
Publication bias	1.896***	−0.804**	2.312***	2.561***	2.293***		(δ) 2.562***
(Standard error)	(0.235)	(0.378)	(0.644)	(0.230)	(0.261)		(0.217)
	[1.434 to 2.357]	[−1.548 to −0.061]	[1.050 to 3.575]	[2.112 to 3.012]	[1.782 to 2.804]		
Effect beyond bias	0.024***	0.088***	0.014	0.012***	0.014***	0.014***	(α1) 0.012***
(Standard error)	(0.008)	(0.010)	(0.016)	(0.002)	(0.002)	(0.001)	(0.001)
	[0.009 to 0.039]	[0.069 to 0.107]	[−0.018 to 0.046]	[0.009 to 0.015]	[0.011 to 0.017]		
Observations	416	416	416	416	416	37	416
Studies	30	30	30	30	30	4	30
Block 2: Decarbonization							
Publication bias	−0.695***	0.353	−0.540	−0.086	−0.534		(δ) −0.086
(Standard error)	(0.262)	(0.270)	(0.663)	(0.310)	(0.332)		(0.417)
	[−1.208 to −0.181]	[−0.180 to 0.887]	[−1.840 to 0.760]	[−0.694 to 0.522]	[−1.185 to 0.117]		
Effect beyond bias	0.001	−0.016***	−0.001	−0.003***	−0.001	−0.003***	(α1) −0.003***
(Standard error)	(0.007)	(0.006)	(0.012)	(0.001)	(0.002)	(0.001)	(0.001)
	[−0.012 to 0.014]	[−0.027 to −0.005]	[−0.025 to 0.023]	[−0.006 to −0.001]	[−0.005 to 0.002]		
Observations	171	171	171	171	171	19	171
Studies	9	9	9	9	9	1	9

Note: The coefficients correspond to publication bias tests applied to subsets of data classified by the dependent variable. Standard errors are in parentheses, and 95% confidence intervals are indicated in brackets. The "publication bias" rows capture the magnitude of the estimated bias associated with the likelihood of publication. The "effect beyond bias" (constant) rows present the average effect corrected for such bias.

***$p < 0.01$.

**$p < 0.05$.

*$p < 0.10$.

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TABLE A9 | Publication Bias Tests for Subsets of Econometric Models.

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Block 1: Method probit/logit model							
Publication bias	4.910***	−5.893	4.417***	3.487***	4.398***		3.487***
(Standard error)	(0.629)	(5.018)	(0.902)	(0.715)	(0.679)		(0.824)
	[3.678 to 6.143]	[−16.020 to 4.234]	[2.649 to 6.186]	[2.086 to 4.889]	[3.067 to 5.728]		
Effect beyond bias	−0.008	0.167**	−0.000	0.006	0.000	0.016***	0.006
(Standard error)	(0.013)	(0.082)	(0.017)	(0.005)	(0.005)	(0.004)	(0.004)
	[−0.034 to 0.018]	[0.002 to 0.333]	[−0.034 to 0.033]	[−0.004 to 0.016]	[−0.010 to 0.010]		
Observations	49	49	49	49	49	12	49
Studies	6	6	6	6	6	1	6
Block 2: Panel Fixed Effects							
Publication bias	0.527**	−1.314*	0.619	1.563***	0.615*		1.563***
(Standard error)	(0.240)	(0.711)	(0.616)	(0.227)	(0.342)		(0.293)
	[0.056 to 0.998]	[−2.714 to 0.086]	[−0.589 to 1.826]	[1.118 to 2.007]	[−0.055 to 1.285]		
Effect beyond bias	0.003	0.035***	0.001	−0.004***	0.002	no observations	−0.004***
(Standard error)	(0.006)	(0.013)	(0.012)	(0.001)	(0.002)		(0.001)
	[−0.009 to 0.015]	[0.010 to 0.060]	[−0.021 to 0.024]	[−0.006 to −0.002]	[−0.003 to 0.006]		
Observations	290	290	290	290	290		290
Studies	15	15	15	15	15		15
Block 3: Ordinary Least Squares(OLS)							
Publication bias	0.292	0.170	1.586*	3.212***	1.584***		3.212***
(Standard error)	(0.335)	(0.332)	(0.859)	(0.416)	(0.319)		(0.466)
	[−0.365 to 0.949]	[−0.487 to 0.827]	[−0.098 to 3.270]	[2.397 to 4.026]	[0.958 to 2.209]		
Effect beyond bias	0.059***	0.063***	0.021	0.012***	0.021***	0.014***	0.012***
(Standard error)	(0.013)	(0.012)	(0.027)	(0.002)	(0.003)	(0.001)	(0.001)
	[0.033 to 0.085]	[0.040 to 0.086]	[−0.033 to 0.074]	[0.009 to 0.015]	[0.015 to 0.026]		
Observations	152	152	152	152	152	14	152
Studies	9	9	9	9	9	1	9

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TABLE A9 | (Continued)

	Panel A (linear)					Panel B (nonlinear)	
	OLS	**FE**	**IV**	**Study**	**Precision**	**WAAP**	**Kink**
Block 4: Instrumental variables							
Publication bias	−0.655	−6.151***	1.026	2.567***	1.024*		2.567***
(Standard error)	(0.523)	(0.682)	(1.235)	(0.365)	(0.579)		(0.400)
	[−1.680 to 0.369]	[−7.515 to −4.787]	[−1.394 to 3.447]	[1.851 to 3.283]	[−0.112 to 2.159]		
Effect beyond bias	0.038***	0.126***	0.011	−0.003***	0.011	−0.002**	−0.003
(Standard error)	(0.010)	(0.011)	(0.021)	(0.001)	(0.007)	(0.000)	(0.002)
	[0.019 to 0.057]	[0.103 to 0.149]	[−0.029 to 0.051]	[−0.005 to −0.001]	[−0.003 to 0.025]		
Observations	73	73	73	73	73	2	73
Studies	12	12	12	12	12	1	12
Block 5: Partial least squares structural equation modeling (PLS-SEM)							
Publication bias	−5.858	−24.991***	5.933	−6.671	−6.259		−80.151
(Standard error)	(3.699)	(2.309)	(5.452)	(4.061)	(3.646)		(187.12)
	[−13.107 to 1.392]	[−29.944–20.038]	[−4.753 to 16.618]	[−14.630 to 1.287]	[−13.405 to 0.887]		
Effect beyond bias	0.663***	1.757***	−0.011	0.709***	0.686***	0.344***	0.348***
(Standard error)	(0.214)	(0.132)	(0.314)	(0.254)	(0.232)	(0.038)	(0.040)
	[0.244 to 1.083]	[1.473 to 2.041]	[−0.626 to 0.605]	[0.211 to 1.207]	[0.231 to 1.142]		
Observations	23	23	23	23	23	23	23
Studies	8	8	8	8	8	8	8

Note: Panel A contains the results with linear specifications, and Panel B applies nonlinear corrections, including precision-based estimations, WAAP, and EK models. The "publication bias" rows capture the magnitude of the estimated bias associated with the likelihood of publication. The "effect beyond bias" (constant) rows present the average effect corrected for such bias. Standard errors are reported in parentheses and 95% confidence intervals in brackets. Some studies use more than one econometric model for their estimates.

***$p < 0.01$.

**$p < 0.05$.

*$p < 0.1$.

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TABLE A10 | Publication Bias Tests for Control Variables.

	Gas emissions	R&D	Environmental regulation	External collaboration	Firm exporting	Ownership type	External financing	Firm growth	Age	Size
Ordinary least squares										
Publication bias	0.799	0.119	0.531	3.778***	1.445***	1.249***	0.435	0.034	1.066***	0.859***
(Standard error)	(2.319)	(0.261)	(0.454)	(0.395)	(0.443)	(0.231)	(0.667)	(0.372)	(0.275)	(0.252)
	[−3.747 to 5.344]	[−0.392 to 0.629]	[−0.359 to 1.420]	[3.005 to 4.551]	[0.577 to 2.314]	[0.795 to 1.702]	[−0.873 to 1.742]	[−0.696 to 0.763]	[0.527 to 1.606]	[0.365 to 1.352]
Effect beyond bias	0.014	0.052***	0.055***	−0.012	0.005	0.021***	0.015**	0.024	0.010	0.020***
(Standard error)	(0.009)	(0.009)	(0.019)	(0.008)	(0.033)	(0.006)	(0.007)	(0.014)	(0.009)	(0.007)
	[−0.004 to 0.032]	[0.035 to 0.069]	[0.018 to 0.091]	[−0.028 to 0.004]	[−0.017 to 0.011]	[0.009 to 0.033]	[0.001 to 0.029]	[−0.04 to 0.052]	[−0.08 to 0.027]	[0.005 to 0.034]
Fixed effects										
Publication bias	0.799	−1.090	−1.956***	−1.745	0.380	−0.594**	0.752	0.145	−0.311	−0.674*
(Standard error)	(2.491)	(0.601)	(0.582)	(3.923)	(2.109)	(0.285)	(2.969)	(0.316)	(0.336)	(0.347)
	[−4.584 to 6.181]	[−2.277 to 0.098]	[−3.139 to −0.773]	[−9.768 to 6.278]	[−3.838 to 4.598]	[−1.154 to −0.034]	[−5.350 to 6.853]	[−0.483 to 0.773]	[−0.971 to 0.349]	[−1.357 to 0.009]
Effect beyond bias	0.014	0.079***	0.148***	0.066	0.010	0.058***	0.012	0.021	0.042***	0.051***
(Standard error)	(0.010)	(0.015)	(0.023)	(0.056)	(0.011)	(0.006)	(0.027)	(0.011)	(0.009)	(0.008)
	[−0.008 to 0.036]	[0.050 to 0.109]	[0.102 to 0.194]	[−0.048 to 0.180]	[−0.011 to 0.031]	[0.046 to 0.071]	[−0.042 to 0.068]	[−0.001 to 0.042]	[0.024 to 0.059]	[0.036 to 0.067]
Observations	15	165	38	33	66	348	29	101	328	409
Studies	1	9	3	3	5	22	2	4	24	28

Note: The reported coefficients correspond to publication bias tests applied to the most commonly used control variables in the literature. Robust standard errors are shown in parentheses, and 95% confidence intervals are reported in brackets. The "publication bias" rows capture the magnitude of the estimated bias associated with the likelihood of publication. The "effect beyond bias" (constant) rows present the average effect corrected for such bias. Many studies report more than one control variable.

***$p < 0.01$.

**$p < 0.05$.

*$p < 0.10$.

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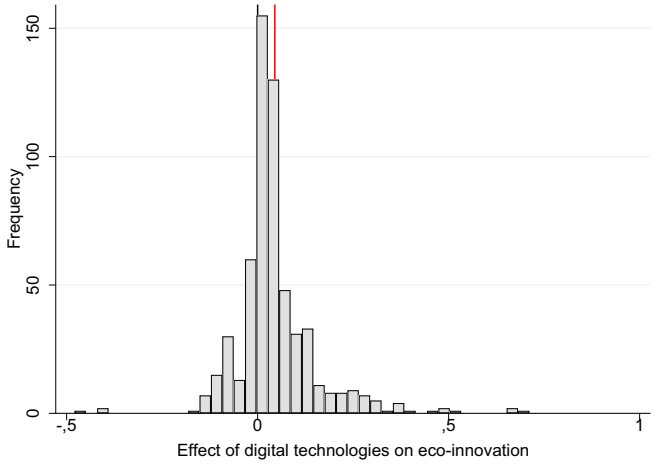


FIGURE A1 | Histogram of estimated effects. This figure presents a histogram of the estimated effects of digital technologies on eco-innovation relative to the number of estimates showing that range of effects; it reveals an asymmetric, positively skewed distribution, with most effects between 0 and 0.5.

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